

ECON6103 Econometrics for Honours

Course Notes

By Xavier Xi Zhang

I use course materials from semester 1, 2025 to make most of the notes and then use 2026's materials to refine the notes. Many of the derivations in Eric's slides are incomplete. I filled in the missing steps and also wrote my own answers to some of the questions. There were also a few minor typos in Eric's slides, and I corrected all of them (thanks to GPT and Teresa).

The core idea of this course:

At the most basic level, data are just numbers collected from the real world through measurement. These numbers may represent inflation, income, productivity, market outcomes, or policy variables, but by themselves they do not provide interpretation. To extract useful information from data, we use probability models, statistical methods, and mathematical tools.

With data and statistical models alone, we can often produce forecasts. For example, using past inflation data, we may predict future inflation. However, forecasting does not necessarily require economic theory, because the statistical model is only detecting patterns in numbers.

The key point is that economic interpretation requires economic theory. Coefficients estimated from probability or regression models do not automatically have economic meaning. A coefficient is not something that exists directly in the real world; it is created within a model. Therefore, to interpret a coefficient as a causal effect, policy effect, treatment effect, or efficiency gain, we must connect the statistical model to an underlying economic model or economic theory.

Economic theory provides concepts such as agents, decision-making, equilibrium, supply and demand, incentives, frictions, market design, and policy mechanisms. These concepts allow us to explain why a statistical relationship can be interpreted as an economically meaningful relationship.

Therefore, the role of econometrics is not simply to run regressions or estimate coefficients. Its real value is in combining data, statistical models, mathematical tools, and economic theory in order to make credible statements about economic phenomena.

Contents

1	Math Review	3
1.1	Probability Theory	3
1.2	Matrix Algebra	7
2	Ordinary Least Squares Estimation	13
2.1	Foundations: Data, Random Variables, and Probability Models	13
2.2	Conditional Moments	15
2.3	Linear Conditional Mean and the Linear Regression Model	16
2.4	The OLS Estimator and Properties	18
2.5	Confidence Intervals in Two Ways	19
2.6	Summary	23
3	Regression Modeling and M-Estimation	24
3.1	Economic Models	25
3.2	The Matchmaking: LR and Economic Theory	25
3.3	BLP (1995, Econometrica): Differentiated Products	26
3.4	The Match Between LR and Linearised BLP	28
3.5	Robustifying the LR Match	29
3.6	M-Estimation	30
3.7	Summary	31
4	The Likelihood Function and Maximum Likelihood Estimation	31
4.1	The Maximum Likelihood Principle	31
4.2	The MLP in Econometrics	32
4.3	Maximum Likelihood Estimator	35
4.3.1	Specification Uncertainty	36
4.3.2	Properties of the MLE	37
4.3.3	Interpretation	38
4.3.4	Alternative Approaches to Specification Uncertainty	38
5	The Bayesian Framework	39
5.1	Econometric Inference and Sources of Information	39
5.2	Bayesian Framework	40
5.3	Bayesian Normal Linear Regression	42

5.4	Monte Carlo Experiment	43
6	Parameter Identification and Instrumental Variable Estimation	43
6.1	Statistical Identification	43
6.2	Structural Identification	45
6.3	Introduction to Instrumental Variables	47
6.4	The Generalized Method of Moments (GMM)	48
7	Inference from Data I: Hypothesis Testing	50
7.1	Hypotheses	50
7.2	Decision Rule and Test Statistics	50
7.3	The Significance Level and Power of the Test	52
7.4	Approximate Hypothesis Tests	53
7.5	P-values	54
7.6	Statistical Significance	54
7.7	Hypothesis Tests vs Confidence Intervals	55
7.8	Joint Hypothesis Tests	56
8	Inference from Data II: Posterior Analysis	56
8.1	Posterior Inference	56
8.2	Probit Example	56
8.3	Bayesian Approach to Specification Uncertainty	58
8.4	Bayesian Model Averaging	58
8.5	Bayesian Hypothesis Testing	59
8.6	Bayesian Tests of Restrictions	60
8.7	Types of Testable Hypotheses	61

1 Math Review

1.1 Probability Theory

Random Variable:

Informally, a **Random Variable (RV)** takes on numerical values determined by an experiment. In fact, it is neither random nor a variable, but rather a function: a mapping from abstract outcomes of a hypothetical experiment to (typically real, but could be complex) numbers.

Example 1.1.1. Consider an experiment in which a fair coin is tossed. Then, the possible outcomes are Head and Tail, i.e., $\{H, T\}$, and we define a random variable X as follows:

$$X = \begin{cases} 0 & \text{if the outcome is } T \\ 1 & \text{if the outcome is } H \end{cases}$$

- X could be either 0 or 1 whenever the coin is tossed.
- Each time, the realisation of X would be different.
- The uncertainty can be summarised as $\Pr(X = 0) = 0.5$.

Probability Functions:

More generally, the uncertainty associated with a random variable (X) is characterised by a probability distribution.

Definition 1.1.2. A probability distribution is represented by its cumulative distribution function (CDF), defined as

$$F_X(x) := \Pr(X \leq x).$$

Definition 1.1.3. When $F_X(x)$ is a continuous function, X is said to be a continuous random variable.

Definition 1.1.4. For a continuous random variable, the probability density function (PDF), denoted by $f_X(x)$, is a function that satisfies

$$\Pr(a < X \leq b) = \int_a^b f_X(x) dx = F_X(b) - F_X(a).$$

If $f_X(x)$ exists and if $F_X(x)$ is differentiable, we have

$$f_X(x) = \frac{d}{dx} F_X(x).$$

If $F_X(x)$ is a step function, X is discrete, which we do not cover today.

Expectation:

A function $g(\cdot)$ of a random variable X is also a random variable.

Example 1.1.5. $g(X) = \exp(X)$, $g(X) = \ln(X)$, etc.

Definition 1.1.6. Suppose X has the PDF $f_X(x)$. The expectation of $g(X)$ is

$$E[g(X)] := \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx.$$

which represents the central tendency of the random variable, $g(X)$.

Definition 1.1.7. As a special case with $g(X) = X$,

$$E[X] := \int_{-\infty}^{\infty} x \cdot f_X(x) dx.$$

which is the expectation of X , or the expected value or the mean.

Definition 1.1.8. Consider another case with $g(X) = (X - E[X])^2$.

$$V(X) := E[(X - E[X])^2] = E[X^2] - E[X]^2,$$

which is called the variance that represents the variability of X . Note that $\sqrt{V(X)}$ is the standard deviation of X .

Property 1.1.9. 1. $E[c] = c$, for any constant c .

2. $E[aX + b] = aE[X] + b$ for any constants a and b .

3. If $a_1, \dots, a_n$ are constants, and $X_1, \dots, X_n$ are RVs, then

$$E\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i E[X_i].$$

As a special case, we have

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i].$$

4. $V(c) = 0$, for any constant c .

5. $V(X) = 0$ if and only if $\Pr(X = c) = 1$ for some constant c .

6. $V(aX + b) = a^2 \text{Var}(X)$ for any constants a and c .

Joint Distribution:

Suppose X and Y are jointly distributed with joint CDF

$$F_{XY}(x, y) = \Pr(X \leq x, Y \leq y).$$

The joint PDF is

$$f_{XY}(x, y) = \frac{d^2}{dx dy} F_{XY}(x, y),$$

if the derivative exists.

The marginal PDF of X can be obtained by integrating out Y ,

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy.$$

X and Y are independent if and only if for all x and y

$$f_{XY}(x, y) = f_X(x)f_Y(y).$$

The expectation of $g(X, Y)$ for some function $g(X, Y)$ is:

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dx dy.$$

Definition 1.1.10. If $g(X, Y) = (X - E[X])(Y - E[Y])$, then we have the covariance,

$$C(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

If $C(X, Y) > 0$, then X and Y “likely move in the same direction”. If $C(X, Y) < 0$, they “likely move in opposite directions”.

$|C(X, Y)| \leq \sqrt{V(X)}\sqrt{V(Y)}$. So, the correlation coefficient

$$\rho_{XY} := \frac{C(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}}$$

is always between -1 and 1 .

If X and Y are independent, $C(X, Y) = 0$, but the converse is generally not true.

Example 1.1.11. $C(X, Y) = 0$ but $C(X^2, Y^2) \neq 0$.

Conditional Distribution:

Suppose X and Y are jointly distributed with the joint PDF $f_{XY}(x, y)$. Then, the conditional PDF of Y given $X = x$ is

$$f_{Y|X}(y|x) := \frac{f_{XY}(x, y)}{f_X(x)},$$

which summarises the distribution of Y when x is the realisation of X .

The conditional CDF is now derived from the conditional PDF as:

$$F_{Y|X}(y|x) = \int_{-\infty}^y f_{Y|X}(y|x) dy.$$

If X and Y are independent, $f_{Y|X}(y|x) = f_Y(y)$ and $f_{X|Y}(x|y) = f_X(x)$.

The conditional expectation of Y given $X = x$ is

$$E[Y|X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy.$$

The conditional variance of Y given $X = x$ is

$$V(Y|X = x) = E[(Y - E[Y|X = x])^2 | X = x].$$

$E[Y|X = x]$ and $V(Y|X = x)$ are functions of x . When x is not specified, $E[Y|X]$ and $V(Y|X)$ are random variables since they are functions of X : i.e., $g(X) \equiv E[Y|X]$ or $g(X) \equiv V[Y|X]$.

Property 1.1.12. *Basic properties of conditional expectations:*

1. $E[c(X)|X] = c(X)$ for any function $c(\cdot)$.
2. $E[a(X)Y + b(X)|X] = a(X)E[Y|X] + b(X)$ for any functions $a(\cdot)$, $b(\cdot)$.
3. If X and Y are independent, $E[Y|X] = E[Y]$.
4. $E[E[Y|X]] = E[Y]$. This is called the law of iterated expectations.
 - More generally, $E[g(X)E[Y|X]] = E[g(X)Y]$.
5. If X and Y are independent, $V(Y|X) = V(Y)$.
6. $E[V(Y|X)] + V(E[Y|X]) = V(Y)$. This is called the law of total variance.

Other useful properties can often be easily derived from the basic properties on the previous slide.

Example 1.1.13. If $E[Y|X] = a_0 + a_1X$ then

$$a_1 = \frac{C(X, Y)}{V(X)} \quad \text{and} \quad a_0 = E[Y] - a_1E[X].$$

Proof. If $E[Y|X] = a_0 + a_1X$ then $E[Y] = a_0 + a_1E[X]$ and $E[XY] = a_0E[X] + a_1E[X^2]$. Now, solve for a_1 to get:

$$a_1 = \frac{E[XY] - E[X]E[Y]}{E[X^2] - E[X]^2} = \frac{C(X, Y)}{V(X)}.$$

□

Normal Distribution:

Suppose a random variable Y follows a normal distribution with mean μ and variance σ^2 . We denote this by:

$$Y \sim \mathcal{N}(\mu, \sigma^2).$$

Y takes on values from $\mathbb{R} = (-\infty, \infty)$.

Every independent draw from $\mathcal{N}(\mu, \sigma^2)$ is a new realisation of Y . The probability that any two draws have exactly the same value is zero!

The uncertainty is summarised by the normal PDF:

$$f_Y(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right).$$

The shape of $f_Y(y|\mu, \sigma^2)$ is given as follows when $\mu = 0$ and $\sigma^2 = 1$ (see Figure 1). It is symmetric about μ and the dispersion is determined by $\sigma := \sqrt{\sigma^2}$. The whole shape is completely determined by (μ, σ^2) .

If $Y \sim \mathcal{N}(\mu, \sigma^2)$ then $E[Y] = \mu$ and $V(Y) = \sigma^2$. All higher-order expectations $E[(Y - E[Y])^k]$, $k = 3, 4, \dots$, exist and are completely determined by $V(Y)$: the normal distribution is the only possible distribution with this feature!

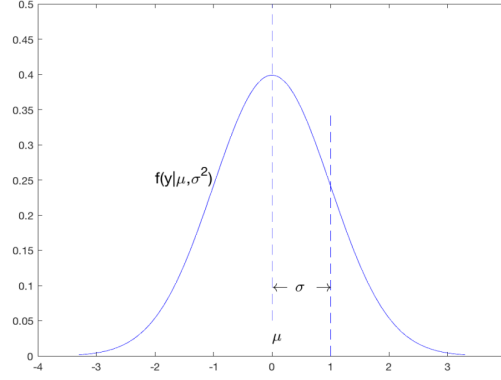


Figure 1: The shape of $f_Y(y|\mu, \sigma^2)$ is given as follows when $\mu = 0$ and $\sigma^2 = 1$.

If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = a_0 + a_1 X$, then

$$Y \sim \mathcal{N}(a_0 + a_1 \mu, a_1^2 \sigma^2).$$

A normal random variable $Y \sim \mathcal{N}(\mu, \sigma^2)$ can always be standardized:

$$X := \frac{Y - \mu}{\sqrt{\sigma^2}}$$

implies $X \sim \mathcal{N}(0, 1)$. $\mathcal{N}(0, 1)$ is called the standard normal distribution, and it plays a crucial role in statistical inference.

PDF and CDF values are easily computed for the standard normal: e.g., if $Y \sim \mathcal{N}(0, 1)$, then

$$F_Y(1.65) = F_Y(1.96) - F_Y(-1.96) = 0.95.$$

1.2 Matrix Algebra

We rarely work with one or two RVs (e.g. X and Y) at a time in econometrics! We use vector and matrix notation to keep things manageable when working with many (often unspecified number of) RVs.

Vectors and matrices are collections of numbers in a certain form. Let

$$\mathbf{a} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad \text{and} \quad \mathbf{b} := \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \quad \text{be } n \text{ dimensional column vectors.}$$

Then,

$$\mathbf{a}' = (a_1 \cdots a_n),$$

the transpose of $\mathbf{a}$, is an n -dim row vector.

Addition is defined when two vectors are in the same form.

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix} \quad \text{and} \quad \mathbf{a}' + \mathbf{b}' = (\mathbf{a} + \mathbf{b})'. \quad \text{But } \mathbf{a}' + \mathbf{b} \text{ is not defined.}$$

Inner product

$$(\mathbf{a}, \mathbf{b}) := \mathbf{a}' \cdot \mathbf{b} = (a_1 \ \cdots \ a_n) \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \sum_{i=1}^n a_i b_i.$$

Consider an $n \times k$ dimensional matrix;

$$\mathbf{A} := \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nk} \end{pmatrix} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_k),$$

where

$$\mathbf{a}_j = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix} \quad \text{for } j = 1, \dots, k.$$

a vector is a special case of a matrix (e.g., $k = 1$).

Transpose:

$$\mathbf{A}' := \begin{pmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{1k} & \cdots & a_{nk} \end{pmatrix} = \begin{pmatrix} \mathbf{a}'_1 \\ \vdots \\ \mathbf{a}'_k \end{pmatrix}.$$

Useful property: $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$.

Addition is allowed between two matrices with the same dimensions. Let

$$\mathbf{B} := \begin{pmatrix} b_{11} & \cdots & b_{1k} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nk} \end{pmatrix} = (\mathbf{b}_1 \ \cdots \ \mathbf{b}_k),$$

where $\mathbf{b}_j$ denotes the j^{th} column of $\mathbf{B}$. Then,

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11} + b_{11} & \cdots & a_{1k} + b_{1k} \\ \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & \cdots & a_{nk} + b_{nk} \end{pmatrix}.$$

But, $\mathbf{A}' + \mathbf{B}$ is not defined unless $k = n$, i.e., $\mathbf{A}$ and $\mathbf{B}$ are square.

To define multiplication, we construct a $k \times m$ matrix

$$\mathbf{C} := \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{k1} & \cdots & c_{km} \end{pmatrix}.$$

Then,

$$\mathbf{A} \times \mathbf{C} := \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nk} \end{pmatrix} \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{k1} & \cdots & c_{km} \end{pmatrix} := \begin{pmatrix} \sum_{\ell=1}^k a_{1\ell} c_{\ell 1} & \sum_{\ell=1}^k a_{1\ell} c_{\ell 2} & \cdots & \sum_{\ell=1}^k a_{1\ell} c_{\ell m} \\ \sum_{\ell=1}^k a_{2\ell} c_{\ell 1} & \sum_{\ell=1}^k a_{2\ell} c_{\ell 2} & \cdots & \sum_{\ell=1}^k a_{2\ell} c_{\ell m} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{\ell=1}^k a_{n\ell} c_{\ell 1} & \sum_{\ell=1}^k a_{n\ell} c_{\ell 2} & \cdots & \sum_{\ell=1}^k a_{n\ell} c_{\ell m} \end{pmatrix}.$$

Another way to view matrix multiplication is to denote $\mathbf{A}$ in terms of its rows and $\mathbf{C}$ in terms of its columns:

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}'_1 \\ \vdots \\ \mathbf{a}'_n \end{pmatrix}, \quad \mathbf{C} = (\mathbf{c}_1 \cdots \mathbf{c}_m),$$

where $\mathbf{a}_i$ is $k \times 1$ and $\mathbf{c}_j$ is $k \times 1$.

Then,

$$\mathbf{A} \times \mathbf{C} = \begin{pmatrix} \mathbf{a}'_1 \mathbf{c}_1 & \cdots & \mathbf{a}'_1 \mathbf{c}_m \\ \vdots & \ddots & \vdots \\ \mathbf{a}'_n \mathbf{c}_1 & \cdots & \mathbf{a}'_n \mathbf{c}_m \end{pmatrix}.$$

The $(i, j)^{\text{th}}$ entry of $\mathbf{A} \times \mathbf{C}$ is

$$\mathbf{a}'_i \mathbf{c}_j = \sum_{\ell=1}^k a_{i\ell} c_{\ell j}.$$

Multiplication is defined only when the number of columns of the first matrix is equal to the number of rows of the second matrix.

Example 1.2.1. For example, $\mathbf{A} \times \mathbf{C}$ is well defined and the resulting matrix is $n \times m$ dimensional but $\mathbf{C} \times \mathbf{A}$ is not defined unless $m = n$.

Suppose $\mathbf{A}$ is a square matrix, i.e., $n = k$.

- a_{ij} is called a diagonal element if $i = j$; otherwise, it is off-diagonal.
- If off-diagonal elements are all zero, $\mathbf{A}$ is said to be a diagonal matrix.
- If $\mathbf{A}' = \mathbf{A}$, it is symmetric ($a_{ij} = a_{ji}$ for all (i, j)).

If $\mathbf{A}$ is diagonal and all diagonal elements are ones, $\mathbf{A}$ is a n -dimensional identity matrix, which we denote by $\mathbf{I}_n$. So,

$$\mathbf{I}_n := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Multiplication with the identity does not change the other matrix. That is,

$$\mathbf{A} \times \mathbf{I}_n = \mathbf{A} \quad \text{and} \quad \mathbf{I}_n \times \mathbf{B} = \mathbf{B}.$$

Essentially, $\mathbf{I}_n$ is the matrix version of the number 1.

For a square $n \times n$ matrix $\mathbf{A}$, if there is another $n \times n$ matrix $\mathbf{B}$ such that

$$\mathbf{I}_n = \mathbf{AB} = \mathbf{BA},$$

then $\mathbf{B}$ is called the inverse of $\mathbf{A}$ and denoted by $\mathbf{A}^{-1}$. In this case, $\mathbf{A}$ is said to be invertible or nonsingular.

1. $(c\mathbf{A})^{-1} = \frac{1}{c}\mathbf{A}^{-1}$ for a scalar c ,
2. $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$,

3. $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$.

If $\mathbf{A}^{-1} = \mathbf{A}'$ then $\mathbf{A}$ is called a unitary matrix.

Every $n \times n$ unitary matrix satisfies

$$\mathbf{A}'\mathbf{A} = \mathbf{A}\mathbf{A}' = \mathbf{I}_n.$$

Essentially, a unitary matrix is the matrix version of ± 1 .

Example 1.2.2.

$$\mathbf{A} = \begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix},$$

where $\mathbf{A} = \mathbf{I}_n$ iff $a = 0$, $\mathbf{A} = -\mathbf{I}_n$ iff $a = \pi$ and

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

iff $a = \pi/2$.

For any $n \times n$ matrix $\mathbf{A}$, the trace of $\mathbf{A}$, $\text{tr}(\mathbf{A})$, is the sum of diagonal elements

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii}.$$

Property 1.2.3. *Properties of the trace:*

1. $\text{tr}(\mathbf{I}_n) = n$,
2. $\text{tr}(\mathbf{A}') = \text{tr}(\mathbf{A})$,
3. $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$,
4. $\text{tr}(c\mathbf{A}) = c \cdot \text{tr}(\mathbf{A})$ for a scalar c ,
5. $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A})$ for any dimension-conformable $\mathbf{A}$ and $\mathbf{B}$.
6. $\text{tr}(\mathbf{A}\mathbf{B}\mathbf{A}') = \text{tr}(\mathbf{A}'\mathbf{B}\mathbf{A}) = \text{tr}(\mathbf{B})$ if $\mathbf{A}$ is unitary.

For any $n \times n$ matrix $\mathbf{A}$, we can also compute the determinant of $\mathbf{A}$, $\det(\mathbf{A}) \equiv |\mathbf{A}|$. For $n = 2$,

$$|\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}.$$

The formula for general n is complicated, but analytically tractable: it can be expressed directly in terms of the elements of $\mathbf{A}$ or recursively using the $n = 2$ formula above.

Property 1.2.4. *Properties of the determinant:*

1. $|\mathbf{A}| = a_{11} \cdots a_{nn}$ if $a_{ij} = 0$ for either all $j > i$ or all $i < j$ (or both),
2. $|\mathbf{A}'| = |\mathbf{A}|$,
3. $|\mathbf{A}|^d = |\mathbf{A}^d|$ for any integer d (incl. negative, so $|\mathbf{A}^{-1}| = |\mathbf{A}|^{-1}$),
4. $|\mathbf{A}| = \pm 1$ if $\mathbf{A}$ is unitary,

5. $|c\mathbf{A}| = c^n \cdot |\mathbf{A}|$ for a scalar c ,
6. $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ if $\mathbf{A}$ and $\mathbf{B}$ are both square,
7. $|\mathbf{A} + \mathbf{B}| \neq |\mathbf{A}| + |\mathbf{B}|$ in general, but
8. $|\mathbf{A} + \mathbf{BDC}'| = |\mathbf{A}||\mathbf{D}||\mathbf{D}^{-1} + \mathbf{C}'\mathbf{A}^{-1}\mathbf{B}|$ if $\mathbf{A}$ and $\mathbf{D}$ are square, but not necessarily the same dimension.

Definition 1.2.5. Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be a set of $n \times 1$ vectors and $c_1, \dots, c_n$ scalars. These are linearly independent if and only if

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots + c_n\mathbf{a}_n = \mathbf{0}$$

implies $c_1 = c_2 = \dots = c_n = 0$.

When $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly independent, $\mathbf{a}_j$ cannot be represented as a linear combination of other $\mathbf{a}_i$, $i \neq j$.

Definition 1.2.6. Let $\mathbf{A}$ be a $n \times k$ matrix. The rank of $\mathbf{A}$, denoted by $\text{rank}(\mathbf{A})$, is the maximum number of linearly independent columns in $\mathbf{A}$.

In general, $\text{rank}(\mathbf{A}) \leq \min\{n, k\}$. If $\text{rank}(\mathbf{A}) = k$, $\mathbf{X}$ has full column rank (which requires $n \geq k$).

Definition 1.2.7. Let $\mathbf{A}$ be a $n \times n$ symmetric matrix and $\mathbf{x}$ a $n \times 1$ vector. The quadratic form of $\mathbf{A}$ is the function $f_{\mathbf{A}}(\mathbf{x}) = \mathbf{x}'\mathbf{A}\mathbf{x}$.

Definition 1.2.8. $\mathbf{A}$ is positive definite (p.d.) if $f_{\mathbf{A}}(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$. $\mathbf{A}$ is positive semi-definite (p.s.d.) if $f_{\mathbf{A}}(\mathbf{x}) \geq 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Property 1.2.9. *Properties of p.d. and p.s.d. matrices:*

1. If $\mathbf{A}$ is p.d., then $a_{ii} > 0$ for all i .
2. If $\mathbf{A}$ is p.s.d., then $a_{ii} \geq 0$ for all i .
3. If $\mathbf{A}$ is p.d., then $\mathbf{A}$ is invertible, i.e., $\mathbf{A}^{-1}$ exists.
4. If $\mathbf{A}$ is $n \times k$, then $\mathbf{A}'\mathbf{A}$ and $\mathbf{A}\mathbf{A}'$ are p.s.d.
5. If $\mathbf{A}$ is $n \times k$ and $\text{rank}(\mathbf{A}) = k$, then $\mathbf{A}'\mathbf{A}$ is p.d. and invertible.

For a given $n \times 1$ vector $\mathbf{a}$, consider the linear function

$$g_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}'\mathbf{x}$$

for all $n \times 1$ vectors $\mathbf{x}$. Then,

$$\frac{\partial g_{\mathbf{a}}(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}.$$

For an $n \times n$ symmetric matrix $\mathbf{A}$, recall the quadratic function

$$f_{\mathbf{A}}(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x}.$$

Then,

$$\frac{\partial f_{\mathbf{A}}(\mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}.$$

Note that

$$\frac{\partial f_{\mathbf{A}}(\mathbf{x})}{\partial \mathbf{x}'} = \left(\frac{\partial f_{\mathbf{A}}(\mathbf{x})}{\partial \mathbf{x}} \right)' \quad \text{and} \quad \frac{\partial g_{\mathbf{a}}(\mathbf{x})}{\partial \mathbf{x}'} = \left(\frac{\partial g_{\mathbf{a}}(\mathbf{x})}{\partial \mathbf{x}} \right)'.$$

Let $\mathbf{y}$ be a $n \times 1$ vector and $\mathbf{X}$ be a $n \times k$ matrix. Consider

$$\min_{\beta_1, \dots, \beta_k} \sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_{ij} \beta_j \right)^2 = \min_{\boldsymbol{\beta}} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Since $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' = \mathbf{y}' - (\mathbf{X}\boldsymbol{\beta})' = \mathbf{y}' - \boldsymbol{\beta}'\mathbf{X}'$, the objective function is

$$\begin{aligned} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) &= (\mathbf{y}' - \boldsymbol{\beta}'\mathbf{X}')(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \end{aligned}$$

Therefore,

$$\min_{\boldsymbol{\beta}} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \min_{\boldsymbol{\beta}} (\mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}).$$

Then, the first order necessary condition (FOC) is:

$$\frac{\partial}{\partial \boldsymbol{\beta}} (\mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}) = -2\frac{\partial}{\partial \boldsymbol{\beta}} \mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}.$$

Applying the rules above,

$$\frac{\partial}{\partial \boldsymbol{\beta}} \mathbf{y}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{y}, \quad \frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta}.$$

The FOC is therefore:

$$-2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0} \iff \mathbf{X}'\mathbf{y} = \mathbf{X}'\mathbf{X}\boldsymbol{\beta},$$

at the optimal solution for $\boldsymbol{\beta}$, say $\hat{\boldsymbol{\beta}}$.

If $\mathbf{X}'\mathbf{X}$ is invertible, the unique solution is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Recall that $\mathbf{X}'\mathbf{X}$ is invertible if and only if $\text{rank}(\mathbf{X}) = k$. What happens if $\text{rank}(\mathbf{X}) < k$? What does it mean?

Suppose we have n random variables, $Y_1, \dots, Y_n$. Then, define the random vector:

$$\mathbf{y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \text{such that} \quad E[\mathbf{y}] = \begin{pmatrix} E[Y_1] \\ \vdots \\ E[Y_n] \end{pmatrix}.$$

Similarly,

$$V(\mathbf{y}) = E[(\mathbf{y} - E[\mathbf{y}])(\mathbf{y} - E[\mathbf{y}])'],$$

where

- The $(i, j)^{\text{th}}$ element of $V(\mathbf{y})$ is $C(Y_i, Y_j)$.
- The i^{th} diagonal element of $V(\mathbf{y})$ is $C(Y_i, Y_i) = V(Y_i)$.
- $V(\mathbf{y})$ is symmetric because $C(Y_i, Y_j) = C(Y_j, Y_i)$.

We can also define a $n \times m$ random matrix consisting of nm random variables $Y_{11}, \dots, Y_{nm}$:

$$\mathbf{Y} = \begin{pmatrix} Y_{11} & \cdots & Y_{1m} \\ \vdots & \ddots & \vdots \\ Y_{n1} & \cdots & Y_{nm} \end{pmatrix},$$

with $E[\mathbf{Y}]$ the $n \times m$ matrix whose $(i, j)^{\text{th}}$ element is $E[Y_{ij}]$. However, $V(\mathbf{Y})$ is not well defined. Let $\mathbf{X}$ be another random matrix (with random vector as a special case). Then, the conditional mean and variance are:

$$E[\mathbf{y} | \mathbf{X}] = \begin{pmatrix} E[Y_1 | \mathbf{X}] \\ \vdots \\ E[Y_n | \mathbf{X}] \end{pmatrix}$$

and

$$V(\mathbf{y} | \mathbf{X}) = E[(\mathbf{y} - E[\mathbf{y} | \mathbf{X}])(\mathbf{y} - E[\mathbf{y} | \mathbf{X}])' | \mathbf{X}].$$

In this case, conditional on $\mathbf{X}$ is short-hand notation for conditional on the RVs $X_{11}, \dots, X_{nm}$. If we let $\mathbb{X}$ denote the set $\{X_{11}, \dots, X_{nm}\}$ and $\mathbf{x}$ denote the $nm \times 1$ random vector $(X_{11}, \dots, X_{nm})'$, then

$$E[\mathbf{y} | \mathbf{X}] \equiv E[\mathbf{y} | \mathbb{X}] \equiv E[\mathbf{y} | \mathbf{x}],$$

$$V(\mathbf{y} | \mathbf{X}) \equiv V(\mathbf{y} | \mathbb{X}) \equiv V(\mathbf{y} | \mathbf{x}).$$

2 Ordinary Least Squares Estimation

2.1 Foundations: Data, Random Variables, and Probability Models

	$\mathbf{y}$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$
1	-1.1185	1	0	0.8328	1.0798	18	-0.1694
2	0.0385	1	0	0.6080	0.8328	16	-0.2481
3	-1.4322	1	1	0.1969	0.6080	15	-1.5417
4	1.8653	1	0	-0.5335	0.1969	11	0.1757
5	-0.6481	1	1	-1.1494	-0.5335	7	-1.4942
6	-0.2756	1	0	-1.7754	-1.1494	5	0.3935
7	-0.4702	1	1	-0.5716	-1.7754	10	0.3012
8	0.9111	1	0	0.7090	-0.5716	2	-0.3986
9	1.2303	1	0	-1.8803	0.7090	4	-1.0730
10	0.6417	1	1	-0.9231	-1.8803	1	0.4680
11	0.2097	1	0	-0.5524	-0.9231	0	0.0477
12	0.9120	1	0	-0.3975	-0.5524	6	0.1169
13	-3.4325	1	0	1.0823	-0.3975	5	-0.1766
14	-0.0056	1	0	-1.2701	1.0823	8	0.2709
15	0.4931	1	1	0.5890	-1.2701	5	-0.5692

Goal: extract useful information from a table of numbers.

The data table has n rows (observations), indexed $i = 1, \dots, n$. In the example, $n = 15$. Each row in the table is an observation, indexed $1, \dots, n$, with $n = 15$. Each column in the table is a $n \times 1$ **vector**.

Assumption 2.1.1. We view each observed column vector as one realisation of a corresponding random vector.

We have arbitrarily labelled the random vectors associated to realisations in each column: $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_k$, with $k = 6$.

Assumption 2.1.2. Assume there exists a joint distribution that characterises all uncertainty associated with $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_k$.

Assuming that observed data was generated by random variables with a joint probability distribution is very general. It is possible to have $\Pr(X_{i1} = X_{j1}) = 1$ for all $i = 1, \dots, n$ and $j = 1, \dots, n$ or $\Pr(X_{i4} = X_{(i-1)3}) = 1$ for all $i = 2, \dots, n$. The only substantive implication: observed data is not the result of chaos (i.e. infinite uncertainty).

To derive practical methods, we need further assumptions about the joint distribution of $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_k$. This is equivalent to constructing a probability model.

- **Intuition:** features of the joint distribution characterise the useful information we aim to extract, along with uncertainty.
- **Advantage:** we can use probability / maths to rigorously analyse the useful information and extraction methods.

Definition 2.1.3 (Probability Model). The sole purpose of a probability model is to describe the joint distribution of $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_k$ in a useful way.

Remark. 1. A probability model is not the true joint distribution (which is itself an assumption) or any other real feature of the world. A probability model is a useful tool for extracting useful info from data. Uncertainty regarding which models are most useful is inherent: continuously evaluate, compare and update.

2. A probability model does not need to be complete: it does not need to rigorously describe every detail of the joint distribution. Useful models focus on features of the distribution related to the information of interest.
3. A probability model is not an economic model. Must be explicitly connected to extract economic information.

Econometric model = a probability model rigorously connected with an economic model.

A useful model of the joint distribution of $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_k$ is often derived from a model of a joint distribution involving additional random variables. There are two ways to do so:

1. Consider the additional random variables $Y_i, X_{i1}, \dots, X_{ik}$ for $i = n + 1, \dots, N$, for some integer $N \geq n$. Let $\tilde{\mathbf{y}}$ denote the $N \times 1$ random vector with elements Y_i , $i = 1, \dots, N$. For each $j = 1, \dots, k$, let $\tilde{\mathbf{x}}_j$ denote the $N \times 1$ random vector with elements X_{ij} , $i = 1, \dots, N$. Note that $\tilde{\mathbf{y}}$ nests $\mathbf{y}$ as a sub-vector and $\tilde{\mathbf{x}}_j$ nests $\mathbf{x}_j$.

2. Similarly, consider the additional random variables $X_{i(k+1)}, \dots, X_{iK}$ for $i = 1, \dots, N$ and $K \geq k$. Using the same notation as above, these define the additional random vectors $\tilde{\mathbf{x}}_{k+1}, \dots, \tilde{\mathbf{x}}_K$.

2.2 Conditional Moments

Let $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_k)$ and $\tilde{\mathbf{X}} = (\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_K)$; hence, $\mathbf{X}$ is a $n \times k$ matrix and $\tilde{\mathbf{X}}$ is a $N \times K$ matrix, which nests $\mathbf{X}$ as its first n rows and first k columns. If $F_{\tilde{\mathbf{y}}, \tilde{\mathbf{X}}}$ is the joint CDF of $\tilde{\mathbf{y}}, \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_K$, then $F_{\mathbf{y}, \mathbf{X}}$ is the implied marginal CDF of $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_k$.

Suppose the purpose is to forecast Y_{n+i} . For any “prediction” $\hat{Y}_{n+i}$, we accept that there is a good chance the prediction will not be perfect. Hence, we define an expected loss:

$$E[(Y_{n+i} - \hat{Y}_{n+i})^2 \mid \tilde{\mathbf{X}}] = E[Y_{n+i}^2 \mid \tilde{\mathbf{X}}] - 2E[Y_{n+i} \mid \tilde{\mathbf{X}}] \hat{Y}_{n+i} + \hat{Y}_{n+i}^2.$$

This expected loss is minimised at $\hat{Y}_{n+i} = E[Y_{n+i} \mid \tilde{\mathbf{X}}]$. Hence, the minimum expected loss is $V(Y_{n+i} \mid \tilde{\mathbf{X}})$.

Table 1: Forecast Scenario

	$\mathbf{y}$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$
1	-1.1185	1	0	0.8328	1.0798	18	-0.1694
2	0.0385	1	0	0.6080	0.8328	16	-0.2481
3	-1.4322	1	1	0.1969	0.6080	15	-1.5417
4	1.8653	1	0	-0.5335	0.1969	11	0.1757
5	-0.6481	1	1	-1.1494	-0.5335	7	-1.4942
6	-0.2756	1	0	-1.7754	-1.1494	5	0.3935
7	-0.4702	1	1	-0.5716	-1.7754	10	0.3012
8	0.9111	1	0	0.7090	-0.5716	2	-0.3986
9	1.2303	1	0	-1.8803	0.7090	4	-1.0730
10	0.6417	1	1	-0.9231	-1.8803	1	0.4680
11	0.2097	1	0	-0.5524	-0.9231	0	0.0477
12	0.9120	1	0	-0.3975	-0.5524	6	0.1169
13	-3.4325	1	0	1.0823	-0.3975	5	-0.1766
14	-0.0056	1	0	-1.2701	1.0823	8	0.2709
15	0.4931	1	1	0.5890	-1.2701	5	-0.5692
16	?	1	0	1.1756	0.4775	3	-1.3012

In the preceding formulation, the features of $F_{\tilde{\mathbf{y}}, \tilde{\mathbf{X}}}$ that stand out are the conditional mean and conditional variance.

- $E[Y_{n+i} \mid \tilde{\mathbf{X}}]$ is the optimal predictor.
- $V(Y_{n+i} \mid \tilde{\mathbf{X}})$ measures uncertainty associated with the optimal predictor.

It makes sense to choose $N = n + 1$, $K = k$ and focus on modelling $E[Y_i \mid \tilde{\mathbf{X}}]$, $V(Y_i \mid \tilde{\mathbf{X}})$, $C(Y_i, Y_j \mid \tilde{\mathbf{X}})$.

Alternatively, suppose the purpose is to assess to what extent X_{ij} helps to predict Y_i for some $i = 1, \dots, n$ and $j = 1, \dots, k$. Let $\mathbf{X}_{-ij}$ denote the set of all elements in $\mathbf{X}$ excluding X_{ij} . We can

evaluate the difference in expected loss:

$$E[(Y_i - E[Y_i | \mathbf{X}_{-ij}])^2 | \mathbf{X}] - V(Y_i | \mathbf{X}) = (E[Y_i | \mathbf{X}] - E[Y_i | \mathbf{X}_{-ij}])^2.$$

In this case it makes sense to set $N = n$, $K = k$ and again focus on modelling $E[Y_i | \mathbf{X}]$, $V(Y_i | \mathbf{X})$, $C(Y_i, Y_j | \mathbf{X})$.

Question: why does "additional random vectors" matter here?

Answer:

1. Forecasting.
2. Asymptotics.
3. Omitted variables.
4. ...

2.3 Linear Conditional Mean and the Linear Regression Model

Recall that $E[Y_i | \tilde{\mathbf{X}}]$ and $V(Y_i | \tilde{\mathbf{X}})$ are functions of the NK random variables $X_{11}, \dots, X_{NK}$.

Without loss of generality (we can always rearrange), we may set $E[Y_i | \tilde{\mathbf{X}}] = E[Y_i | \tilde{\mathbf{x}}_i]$, where $\tilde{\mathbf{x}}_i = (X_{i1}, \dots, X_{iK})$ is the i th row of $\tilde{\mathbf{X}}$.

With only a small loss of generality (Taylor Expansion), we may assume that $E[Y_i | \tilde{\mathbf{x}}_i]$ is a linear function: There exist quantities $\alpha_{i1}, \dots, \alpha_{iK}$ such that

$$E[Y_i | \tilde{\mathbf{x}}_i] = \sum_{j=1}^K X_{ij} \alpha_{ij} = \tilde{\mathbf{x}}_i' \tilde{\boldsymbol{\alpha}}_i,$$

where $\tilde{\boldsymbol{\alpha}}_i = (\alpha_{i1}, \dots, \alpha_{iK})$ is a $K \times 1$ **vector**. $\tilde{\boldsymbol{\alpha}}_i$ does not depend on any element of $\tilde{\mathbf{x}}_i$.

Let $\mathbf{x}_i = (X_{i1}, \dots, X_{ik})'$ denote the observed k regressors (we don't observe extra regressors). Then

$$E[Y_i | \mathbf{x}_i] = E\left(\mathbf{x}_i' \tilde{\boldsymbol{\alpha}}_i + \sum_{j=k+1}^K X_{ij} \alpha_{ij} \middle| \mathbf{x}_i\right) = \mathbf{x}_i' \tilde{\boldsymbol{\alpha}}_i + \sum_{j=k+1}^K E[X_{ij} | \mathbf{x}_i] \alpha_{ij}.$$

A **stronger assumption** imposes that $E[Y_i | \mathbf{x}_i]$ is linear:

$$E[Y_i | \mathbf{x}_i] = \mathbf{x}_i' \boldsymbol{\beta}_i,$$

which requires either $E[X_{ij} | \mathbf{x}_i]$ is linear in $\mathbf{x}_i$ or $\alpha_{ij} = 0$ for $j = k + 1, \dots, K$.

Since $E[\mathbf{x}_i E[Y_i | \mathbf{x}_i]] = E[\mathbf{x}_i Y_i]$, linearity implies the **moment equation**

$$E[\mathbf{x}_i Y_i] = E[\mathbf{x}_i \mathbf{x}_i'] \boldsymbol{\beta}_i.$$

$\boldsymbol{\beta}_i$ is uniquely determined iff $\text{rank } E[\mathbf{x}_i \mathbf{x}_i'] = k$.

- If $\text{rank } E[\mathbf{x}_i \mathbf{x}_i'] = k$, then

$$E[Y_i | \mathbf{x}_i] = \mathbf{x}_i' \boldsymbol{\beta}_i = \mathbf{x}_i' E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_i Y_i].$$

Note that $E[\mathbf{x}_i \mathbf{x}_i']$ is a matrix, where elements are all integral.

- If $\text{rank } E[\mathbf{x}_i \mathbf{x}_i'] < k$, then $E[\mathbf{x}_i \mathbf{x}_i']$ is not positive definite., and there exists $\mathbf{c}_i \neq \mathbf{0}$ such that

$$\mathbf{c}_i' E[\mathbf{x}_i \mathbf{x}_i'] \mathbf{c}_i = 0 \Rightarrow E[(\mathbf{c}_i' \mathbf{x}_i)^2] = 0 \Rightarrow \mathbf{c}_i' \mathbf{x}_i = \sum_{j=1}^k X_{ij} c_{ij} = 0 \text{ with probability 1.}$$

Hence some X_{ij} are redundant: linear functions of other elements of $\mathbf{x}_i$. In other words, they are linear dependent!

Assumption 2.3.1 (A1 + A2).

$$(i) E[Y_i | \mathbf{x}_i] \text{ is linear,} \quad (ii) \text{rank } E[\mathbf{x}_i \mathbf{x}_i'] = k, \quad i = 1, \dots, n.$$

Remark. (i) “Strong” does not mean “wrong”; it means the assumption should be reassessed in the context of the application. Often, relaxing it does not lead to a better practical method.

Assumption 2.3.2 (A3). Assume $E[\mathbf{x}_i \mathbf{x}_i']$ and $E[\mathbf{x}_i Y_i]$ are the same for $i = 1, \dots, n$. That is

$$\mathbf{M}_x = E[\mathbf{x}_i \mathbf{x}_i'], \quad \mathbf{m}_{y,x} = E[\mathbf{x}_i Y_i], \quad \boldsymbol{\beta}_i = \mathbf{M}_x^{-1} \mathbf{m}_{y,x} \equiv \boldsymbol{\beta}.$$

Then $E[Y_i | \mathbf{x}_i]$ is determined by the $k(k+3)/2$ unconditional moments in $E[\mathbf{x}_i \mathbf{x}_i']$ and $E[\mathbf{x}_i Y_i]$.

However, we cannot compute these moments directly from the observed data: they are integrals involving $F_{y,X}$. We need a link from model objects (expectations) to sample objects.

One way (of many) to establish a link between the probability model we have constructed and the data is to define

Definition 2.3.3 (Ordinary Least Squares (OLS) Estimator).

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{y} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i \right),$$

where $\hat{\boldsymbol{\beta}}$ is called the Ordinary Least Squares estimator and is a $k \times 1$ random vector because it is a function the random variables Y_i and X_{ij} , $i = 1, \dots, n$, $j = 1, \dots, k$.

Remark. (i) $\hat{\boldsymbol{\beta}}$ is only a function of RVs for which we have observed one realisation: we can plug in the data in place of $\mathbf{X}$ and $\mathbf{y}$ to obtain one realisation of $\hat{\boldsymbol{\beta}}$. However, $\boldsymbol{\beta}$ has no uncertainty; it is not a RV,

(ii) Note the connections: $\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \approx E[\mathbf{x}_i \mathbf{x}_i']$, $\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i \approx E[\mathbf{x}_i Y_i]$.

OLS solves the least squares problem:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b}} \frac{1}{n} (\mathbf{y} - \mathbf{X} \mathbf{b})' (\mathbf{y} - \mathbf{X} \mathbf{b}) = \arg \min_{\mathbf{b}} \frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \mathbf{b})^2.$$

This connects to the conditional expected loss viewpoint:

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \mathbf{b})^2 \approx E[(Y_i - \mathbf{x}_i' \mathbf{b})^2 | \mathbf{X}].$$

Under the linearity (A1), rank (A2), and invariant moments (A3) assumptions:

$$E[\hat{\boldsymbol{\beta}} | \mathbf{X}] = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' E[\mathbf{y} | \mathbf{X}] = (\mathbf{X}' \mathbf{X})^{-1} (\mathbf{X}' \mathbf{X}) \boldsymbol{\beta} = \boldsymbol{\beta} = E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_i Y_i] = \mathbf{M}_x^{-1} \mathbf{m}_{y,x}.$$

2.4 The OLS Estimator and Properties

Thus OLS is **unbiased** (and mean-independent of $\mathbf{X}$ in the lecture's phrasing).

Theorem 2.4.1 (Unbiasedness).

$$E[\hat{\beta}] = \beta = E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_i Y_i].$$

Example 2.4.2 (Intercept + one regressor). Let $k = 2$ with $X_{i1} = 1$ for all i . Then

$$\beta_2 = \frac{C(X_{i2}, Y_i)}{V(X_{i2})}.$$

In this case, the reduction in expected loss from using X_{i2} is

$$\left(E[Y_i | X_{i1}] - E[Y_i | X_{i1}, X_{i2}]\right)^2 = (E[Y_i] - E[Y_i | X_{i2}])^2 = \beta_2^2 (X_{i2} - E[X_{i2}])^2.$$

So β_2^2 measures how much X_{i2} reduces expected loss per squared deviation from its mean. The OLS slope is

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (X_{i2} - \bar{X}_2)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_{i2} - \bar{X}_2)^2}.$$

Moreover,

$$E[\hat{\beta}_2^2 | \mathbf{X}] = V(\hat{\beta}_2 | \mathbf{X}) + \beta_2^2 = \frac{\sum_{i=1}^n \sum_{j=1}^n (X_{i2} - \bar{X}_2)(X_{j2} - \bar{X}_2) C(Y_i, Y_j | \mathbf{X})}{\left(\sum_{i=1}^n (X_{i2} - \bar{X}_2)^2\right)^2} + \beta_2^2.$$

More generally, $V(\hat{\beta} | \mathbf{X}) = E[\hat{\beta} \hat{\beta}' | \mathbf{X}] - \beta \beta'$, and

$$\left(E[Y_i] - E[Y_i | \mathbf{x}_i]\right)^2 = \text{tr}\left(\beta \beta' (\mathbf{x}_i - E[\mathbf{x}_i])(\mathbf{x}_i - E[\mathbf{x}_i])'\right).$$

$V(\hat{\beta} | \mathbf{X})$ measures the conditional bias of $\hat{\beta} \hat{\beta}'$ as an estimator of $\beta \beta'$.

$V(\hat{\beta} | \mathbf{X})$ depends on $V(Y_i | \mathbf{X})$ and $C(Y_i, Y_j | \mathbf{X})$:

$$V(\hat{\beta} | \mathbf{X}) = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' V(\mathbf{y} | \mathbf{X}) \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1}.$$

Recall from ECON3330: $V(\mathbf{y} | \mathbf{X})$ is called covariance matrix.

Assumption 2.4.3 (A4 + A5). $C(Y_i, Y_j | \mathbf{X}) = 0$ and $V(Y_i | \mathbf{X}) = \sigma_y^2$ for all $i = 1, \dots, n$, $j \neq i$; equivalently $V(\mathbf{y} | \mathbf{X}) = \sigma_y^2 \mathbf{I}_n$.

$C(Y_i, Y_j | \mathbf{X}) = E[Y_i Y_j | \mathbf{X}] - \beta' \mathbf{x}_i \mathbf{x}_j' \beta = 0$ implies

$$E[Y_i Y_j] = E[\mathbf{x}_i Y_i]' E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_i \mathbf{x}_j'] E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_j Y_j].$$

$V(Y_i | \mathbf{X}) = E[Y_i^2 | \mathbf{X}] - \beta' \mathbf{x}_i \mathbf{x}_i' \beta = \sigma_y^2$ implies

$$\sigma_y^2 = E[Y_i^2] - E[\mathbf{x}_i Y_i]' E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_i Y_i].$$

Under the invariant moments ($E[\mathbf{x}_i Y_i]$ and $E[\mathbf{x}_i \mathbf{x}_i']$) assumption, this implies $E[Y_i^2]$ and $E[Y_i Y_j]$ are also invariant.

$V(\mathbf{y} \mid \mathbf{X}) = \sigma_y^2 \mathbf{I}_n$ implies

$$V(\hat{\beta} \mid \mathbf{X}) = \sigma_y^2 (\mathbf{X}' \mathbf{X})^{-1}.$$

Theorem 2.4.4 (Gauss–Markov Theorem). *Let $\tilde{\beta} = \mathbf{W}\mathbf{y}$ be an alternative linear, unbiased estimator of β . Assume $V(\mathbf{y} \mid \mathbf{X}) = \sigma_y^2 \mathbf{I}_n$. If the linearity, rank, and invariance assumptions also hold, then*

$$V(\tilde{\beta} \mid \mathbf{X}) - V(\hat{\beta} \mid \mathbf{X}) \text{ is positive semi-definite.}$$

This is just for fun...

Proof. Let $\mathbf{A} = \mathbf{W} - (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$ (a $k \times n$ matrix). Then $\tilde{\beta} = \hat{\beta} + \mathbf{A}\mathbf{y}$. Moreover,

$$E[\tilde{\beta} \mid \mathbf{X}] = E[\hat{\beta} \mid \mathbf{X}] + \mathbf{A}E[\mathbf{y} \mid \mathbf{X}] = \beta + \mathbf{A}\mathbf{X}\beta.$$

Thus $\tilde{\beta}$ is unbiased iff $\mathbf{A}\mathbf{X} = \mathbf{0}$. Then

$$\begin{aligned} V(\tilde{\beta} \mid \mathbf{X}) &= ((\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' + \mathbf{A}) V(\mathbf{y} \mid \mathbf{X}) ((\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' + \mathbf{A})' \\ &= \sigma_y^2 \left[(\mathbf{X}' \mathbf{X})^{-1} + \mathbf{A}\mathbf{A}' + \mathbf{A}\mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} + (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{A}' \right] \\ &= V(\hat{\beta} \mid \mathbf{X}) + \sigma_y^2 \mathbf{A}\mathbf{A}', \end{aligned}$$

where the cross terms vanish because $\mathbf{A}\mathbf{X} = \mathbf{0}$. Since $\mathbf{A}\mathbf{A}'$ is p.s.d., the claim follows. $\square$

2.5 Confidence Intervals in Two Ways

Gauss–Markov Theorem is a neat result, but how does $\hat{\beta}_j$ help us measure how much X_{ij} reduces expected loss? Given the preceding assumptions, and setting $\mathbf{S} = (\mathbf{X}' \mathbf{X})^{-1}$, we have $V(\hat{\beta}_j \mid \mathbf{X}) = \sigma_y^2 S_{jj}$. The standard error is: $se(\hat{\beta}_j) = \sqrt{\sigma_y^2 S_{jj}}$.

We may now ask: How many standard errors below β_j would constitute a very unlikely realisation of $\hat{\beta}_j$? That is: we want to measure the uncertainty. Formally, we:

1. quantify “very unlikely” with a small probability α (e.g. $\alpha = 0.025$ or $\alpha = 0.05$);
2. formulate the probability statement:

$$\Pr(\hat{\beta}_j \leq \beta_j - z_l se(\hat{\beta}_j)) = \alpha;$$

3. invert the equation to solve for z_l as a function of α .

Similarly, we may ask: How many standard errors above β_j would constitute a very unlikely realisation of $\hat{\beta}_j$? compute z_u from $\Pr(\hat{\beta}_j \geq \beta_j + z_u se(\hat{\beta}_j)) = \alpha$.

Combining very unlikely small with very unlikely large yields:

$$\Pr(\hat{\beta}_j \leq \beta_j - z_l se(\hat{\beta}_j) \text{ or } \hat{\beta}_j \geq \beta_j + z_u se(\hat{\beta}_j)) = 2\alpha,$$

since the two events are mutually exclusive.

This implies that realisations of $\hat{\beta}_j$ will most likely be within z_l std errors below β_j and within z_u std errors above β_j :

$$\Pr(\beta_j - z_l se(\hat{\beta}_j) \leq \hat{\beta}_j \leq \beta_j + z_u se(\hat{\beta}_j)) = 1 - 2\alpha.$$

It is often interpreted as follows.

Definition 2.5.1. Consider the interval $CI \equiv [\hat{\beta}_j - z_u se(\hat{\beta}_j), \hat{\beta}_j + z_l se(\hat{\beta}_j)]$. The probability that the lower bound $\hat{\beta}_j - z_u se(\hat{\beta}_j)$ is below β_j and simultaneously the upper bound $\hat{\beta}_j + z_l se(\hat{\beta}_j)$ is above β_j , is precisely $1 - 2\alpha$. CI is called the $(1 - 2\alpha) \times 100\%$ **confidence interval** for β_j .

If $\alpha = 0.025$, we may conclude the random interval CI (since the lower and upper bounds are random variables) contains the deterministic quantity β_j with 95% probability.

Confidence intervals can often be easily transformed. Let $\hat{l}_j \equiv \hat{\beta}_j - z_u se(\hat{\beta}_j)$ and $\hat{u}_j \equiv \hat{\beta}_j + z_l se(\hat{\beta}_j)$; $\hat{l}_j \leq \beta_j \leq \hat{u}_j$ implies:

$$\begin{aligned} \hat{l}_j^2 &\leq \beta_j^2 \leq \hat{u}_j^2 && \text{if } \hat{l}_j \geq 0, \\ \hat{u}_j^2 &\leq \beta_j^2 \leq \hat{l}_j^2 && \text{if } \hat{u}_j \leq 0, \\ 0 &\leq \beta_j^2 \leq \max\{|\hat{l}_j|, |\hat{u}_j|\}^2 && \text{if } \hat{l}_j < 0 < \hat{u}_j. \end{aligned}$$

Consequently, an interval formed using the bounds defined above will have confidence level at least 95%.

Question: How much does X_{ij} reduce expected loss?

Answer:

1. $\hat{\beta}$ can be labeled "point estimate" but that doesn't give it any useful interpretation in itself.
2. The answer comes entirely from the CI. Say the 95% CI is $[0.1, 1.5]$. Then the answer is, "the reduction in expected loss is between 0.1 and 1.5 with 95% confidence"

Remark. All values outside the interval are equally unreasonable; all values inside the interval are equally reasonable. **$\hat{\beta}$ can only help us construct the CI; nothing else. It has no other interpretation.**

To actually compute CIs in practice, we need $se(\hat{\beta}_j)$, z_l and z_u . $se(\hat{\beta}_j)$ depends on σ_y^2 , which we have so far left unspecified.

Recall that $\sigma_y^2 = E[Y_i^2] - E[\mathbf{x}_i Y_i]' E[\mathbf{x}_i \mathbf{x}_i']^{-1} E[\mathbf{x}_i Y_i]$. As with $\hat{\beta}$, we can try "plugging in" sample moments for all the expectations in the expression above:

$$\hat{\sigma}_y^2 = \frac{1}{n} \mathbf{y}' \left(\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \mathbf{y} = \frac{1}{n} \sum_i Y_i^2 - \left(\frac{1}{n} \mathbf{x}_i Y_i \right)' \left(\frac{1}{n} \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\frac{1}{n} \mathbf{x}_i Y_i \right).$$

Theorem 2.5.2 (Mean-Independence of $\hat{\sigma}_y^2$ and $\mathbf{X}$). Set $\mathbf{M}_x = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'$ and $\hat{\mathbf{u}} = \mathbf{M}_x \mathbf{y}$. Then, $\hat{\sigma}_y^2 = \frac{1}{n} \hat{\mathbf{u}}' \hat{\mathbf{u}}$, $E[\hat{\sigma}_y^2 | \mathbf{X}] = \frac{n-k}{n} \sigma_y^2$, $C(\hat{\beta} - \beta, \hat{\mathbf{u}} | \mathbf{X}) = 0$.

Proof. It can be easily verified that $\mathbf{M}_x^2 = \mathbf{M}_x$, $\text{tr } \mathbf{M}_x = n - k$ and $\mathbf{M}_x \mathbf{X} = 0$. It follows immediately that $\hat{\sigma}_y^2 = \frac{1}{n} \hat{\mathbf{u}}' \hat{\mathbf{u}}$,

$$E[\hat{\sigma}_y^2 | \mathbf{X}] = \frac{1}{n} E[\mathbf{y}' \mathbf{M}_x \mathbf{y} | \mathbf{X}] = \frac{1}{n} E[\text{tr}(\mathbf{y}' \mathbf{M}_x \mathbf{y}) | \mathbf{X}] = \frac{1}{n} E[\text{tr}(\mathbf{M}_x \mathbf{y} \mathbf{y}') | \mathbf{X}] = \frac{1}{n} \text{tr}(\mathbf{M}_x E[\mathbf{y} \mathbf{y}' | \mathbf{X}])$$

$$= \frac{1}{n} \text{tr} \left(\mathbf{M}_x \left(V(\mathbf{y} \mid \mathbf{X}) + E[\mathbf{y} \mid \mathbf{X}] E[\mathbf{y} \mid \mathbf{X}]' \right) \right) = \frac{\text{tr} \mathbf{M}_x}{n} \sigma_y^2 + \frac{1}{n} \boldsymbol{\beta}' \mathbf{X}' \mathbf{M}_x \mathbf{X} \boldsymbol{\beta} = \frac{n-k}{n} \sigma_y^2,$$

and

$$\begin{aligned} C(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \hat{\mathbf{u}} \mid \mathbf{X}) &= E[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \hat{\mathbf{u}}' \mid \mathbf{X}] - E[\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \mid \mathbf{X}] E[\hat{\mathbf{u}}' \mid \mathbf{X}] \\ &= (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' E[\mathbf{y} \mathbf{y}' \mid \mathbf{X}] \mathbf{M}_x - \boldsymbol{\beta} E[\mathbf{y} \mid \mathbf{X}]' \mathbf{M}_x = \sigma_y^2 (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{M}_x. \end{aligned}$$

□

The important conclusion is that $\hat{\sigma}_y^2$ is mean-independent of $\mathbf{X}$. It can be easily made unbiased by re-scaling:

$$E \left[\frac{n}{n-k} \hat{\sigma}_y^2 \mid \mathbf{X} \right] = \sigma_y^2.$$

We may use $\hat{\sigma}_y^2$ instead of σ_y^2 in $\widehat{se}(\hat{\beta}_j) = \sqrt{\hat{\sigma}_y^2 S_{jj}}$ and:

$$\Pr(\hat{\beta}_j \leq \beta_j - z_l \widehat{se}(\hat{\beta}_j)) = \alpha, \quad \Pr(\hat{\beta}_j \geq \beta_j + z_u \widehat{se}(\hat{\beta}_j)) = \alpha.$$

However, this is still not enough: to obtain z_l and z_u from α .

- We need the CDF for the RV $\hat{z} \equiv (\hat{\beta}_j - \beta_j) / \widehat{se}(\hat{\beta}_j)$.
- Hence, we need to at least specify the conditional distribution $(\mathbf{y} \mid \mathbf{X})$.

Assumption 2.5.3 (A6 (i)). $(\mathbf{y} \mid \mathbf{X}) \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma_y^2 \mathbf{I}_n)$.

This assumption implies $E(Y_i \mid \mathbf{x}_i) = \mathbf{x}_i' \boldsymbol{\beta}$ and $V(\mathbf{y} \mid \mathbf{X}) = \sigma_y^2 \mathbf{I}_n$. In fact, Y_i and Y_j are conditionally independent. This turns out to be the only practically useful choice to obtain an exact distribution (and, therefore, exact CDF) for $\hat{z}$.

From conditional normality, we obtain:

- $(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) \sim \mathcal{N}(\boldsymbol{\beta}, \sigma_y^2 (\mathbf{X}' \mathbf{X})^{-1})$, which implies $(\hat{\beta}_j - \beta_j) / se(\hat{\beta}_j) \sim \mathcal{N}(0, 1)$.
- $\hat{\mathbf{u}}' \hat{\mathbf{u}} / \sigma_y^2 \sim \chi_{n-k}^2$ (Chi-squared distribution).
- $C(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \hat{\mathbf{u}} \mid \mathbf{X}) = 0$ implies $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\hat{\mathbf{u}}$ are independent.
 - $(\hat{\beta}_j - \beta_j) / se(\hat{\beta}_j)$ and $\hat{\sigma}_y^2$ are independent.
 - $\sqrt{\frac{n-k}{n}} (\hat{\beta}_j - \beta_j) / \widehat{se}(\hat{\beta}_j) \sim t_{n-k}$ (Student- t distribution).

Therefore, for any z ,

$$\Pr(\hat{\beta}_j \geq \beta_j + z \widehat{se}(\hat{\beta}_j)) = \Pr(\hat{\beta}_j \leq \beta_j - z \widehat{se}(\hat{\beta}_j)) = 1 - \Pr \left(\sqrt{\frac{n-k}{n}} \frac{\hat{\beta}_j - \beta_j}{\widehat{se}(\hat{\beta}_j)} \leq z \sqrt{\frac{n-k}{n}} \right),$$

which yields $z = z_l = z_u$ and $t = z \sqrt{(n-k)/n}$ is computed as the $(1 - \alpha)$ th percentile of the Student- t distribution; the $(1 - 2\alpha) \times 100\%$ CI is simply

$$\hat{\beta}_j \pm t \sqrt{\frac{n}{n-k}} \widehat{se}(\hat{\beta}_j).$$

If we do not like the assumption $(\mathbf{y} \mid \mathbf{X}) \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma_y^2 \mathbf{I}_n)$, we can replace it with alternative assumptions.

Let us keep k fixed but consider additional RVs, Y_i and $\mathbf{x}_i$ such that the OLS estimator and the variance estimator are:

$$\tilde{\boldsymbol{\beta}} = (\tilde{\mathbf{X}}' \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}' \tilde{\mathbf{y}}, \quad \tilde{\sigma}_y^2 = \frac{1}{N} \tilde{\mathbf{y}}' \left(\mathbf{I}_k - \tilde{\mathbf{X}} (\tilde{\mathbf{X}}' \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}' \right) \tilde{\mathbf{y}}.$$

All assumptions except the Normal distribution assumption are extended to the additional random variables.

To replace the Normal distribution assumption, instead let $U_i \equiv Y_i - E[Y_i \mid \mathbf{x}_i] = Y_i - \mathbf{x}_i' \boldsymbol{\beta}$; $\tilde{\mathbf{u}} = (U_1, \dots, U_N)'$.

Assumption 2.5.4 (A6 (ii)). Assume $(\mathbf{x}_i, U_i)$ and $(\mathbf{x}_j, U_j)$ are independent for all $i \neq j$.

The independence assumption (along with the other assumptions) implies:

- $\tilde{\mathbf{X}}' \tilde{\mathbf{X}} / N \xrightarrow{p} E[\mathbf{x}_i \mathbf{x}_i'] \equiv \mathbf{Q}$ as $N \rightarrow \infty$ (WLLN);
- $\tilde{\mathbf{X}}' \tilde{\mathbf{u}} / N \xrightarrow{p} E[\mathbf{x}_i U_i] = \mathbf{0}$ as $N \rightarrow \infty$ (WLLN);
- $\tilde{\mathbf{u}}' \tilde{\mathbf{u}} / N \xrightarrow{p} E[U_i^2] = \sigma^2$ as $N \rightarrow \infty$ (WLLN);
- Consequently,

$$\tilde{\sigma}_y^2 = \frac{\tilde{\mathbf{u}}' \tilde{\mathbf{M}}_x \tilde{\mathbf{u}}}{N} = \frac{\tilde{\mathbf{u}}' \tilde{\mathbf{u}}}{N} - \left(\frac{\tilde{\mathbf{X}}' \tilde{\mathbf{u}}}{N} \right)' \left(\frac{\tilde{\mathbf{X}}' \tilde{\mathbf{X}}}{N} \right)^{-1} \left(\frac{\tilde{\mathbf{X}}' \tilde{\mathbf{u}}}{N} \right) \xrightarrow{p} \sigma_y^2.$$

- $\tilde{\mathbf{X}}' \tilde{\mathbf{u}} / \sqrt{N} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{Q})$ as $N \rightarrow \infty$ (Lindberg-Feller CLT);
- $\sqrt{N}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left(\frac{\tilde{\mathbf{X}}' \tilde{\mathbf{X}}}{N} \right)^{-1} \left(\frac{\tilde{\mathbf{X}}' \tilde{\mathbf{u}}}{\sqrt{N}} \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{Q}^{-1})$ as $N \rightarrow \infty$ (Slutsky Lemma).

Result: Letting $\tilde{\mathbf{S}} = (\tilde{\mathbf{X}}' \tilde{\mathbf{X}})^{-1}$ and $\widehat{se}(\tilde{\beta}_j) = \sqrt{\tilde{\sigma}_y^2 \tilde{S}_{jj}}$, we have $N \tilde{\mathbf{S}} \xrightarrow{p} \mathbf{Q}^{-1}$ and

$$\frac{\tilde{\beta}_j - \beta_j}{\widehat{se}(\tilde{\beta}_j)} = \frac{\sqrt{N}(\tilde{\beta}_j - \beta_j)}{\sqrt{\tilde{\sigma}_y^2 (N \tilde{S}_{jj})}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Putting everything all together

- If $(\mathbf{y} \mid \mathbf{X}) \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma_y^2 \mathbf{I}_n)$ (or $(\mathbf{u} \mid \mathbf{X}) \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I}_n)$) and the other assumptions hold for $i = 1, \dots, n$, then we can construct **exact** $(1 - 2\alpha) \times 100\%$ CIs using $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}_y^2$ and the Student- t CDF.
- If $(\mathbf{x}_i, U_i)$ are independent and satisfy the other assumptions for $i = 1, \dots, N$, we can hypothetically construct CIs using $\tilde{\boldsymbol{\beta}}$, $\tilde{\sigma}_y^2$ and the standard Normal CDF:
 - CI significance level is approximately $(1 - 2\alpha) \times 100\%$,
 - approximation error goes to 0 as $N \rightarrow \infty$,
 - Student- t with $N - k$ d.f. converges to standard Normal as $N \rightarrow \infty$ (with k fixed),

- in practice we never observe $N > n$; we use $\hat{\beta}, \hat{\sigma}_y^2$ to approximate the asymptotic CI. These are **asymptotic CIs**.

We compute

$$CI : \hat{\beta}_j \pm z \widehat{se}(\hat{\beta}_j),$$

regardless of whether or not normality is assumed. The only difference is how we set z :

- **Normality assumption:** set $z = t \sqrt{\frac{n}{n-k}}$, where t is the $(1 - \alpha)$ th percentile of the Student- t CDF.
- **Alternative assumption:** set z as the $(1 - \alpha)$ th percentile of the standard Normal CDF.

Question: When is the difference more important or less important?

Answer:

1. Numerical difference: driven largely by degrees of freedom $n - k$.
2. The normality assumption only helps if the other assumptions also hold: if $V(Y_i | \mathbf{x}_i) \neq V(Y_j | \mathbf{x}_j)$ or $C(Y_i, Y_j | \mathbf{X}) \neq 0$ for some $i \neq j$, then normality does not rescue classical CI formulas.

2.6 Summary

To extract information from a given set of data using statistics tools, we always begin by assuming the data is a realisation of some RVs. We assumed the relevant information is contained the conditional mean; we focused on inference from $E[Y_i | \mathbf{x}_i]$ based on confidence intervals.

We developed inference from $E[Y_i | \mathbf{x}_i]$ based on the following assumptions:

- **A1 (linearity):** $E[Y_i | \mathbf{x}_i]$ is a linear function for $i = 1, \dots, n$.
- **A2 (full rank):** $\text{rank } E[\mathbf{x}_i \mathbf{x}_i'] = k$ for $i = 1, \dots, n$.
- **A3 (invariant moments):** $E[\mathbf{x}_i \mathbf{x}_i']$ and $E[\mathbf{x}_i Y_i]$ are constant for $i = 1, \dots, n$.
- **A4 (no correlation):** $C(Y_i, Y_j | \mathbf{X}) = 0$ for all $i \neq j$, $i = 1, \dots, n$, $j = 1, \dots, n$.
- **A5 (homoscedasticity):** $V(Y_i | \mathbf{X})$ is constant for $i = 1, \dots, n$.
- **A6(i) (normality):** $(\mathbf{y} | \mathbf{X}) \sim N(\mathbf{X}\beta, \sigma_y^2 \mathbf{I}_n)$, or
- **A6(ii) (independence):** A1–A5 are satisfied for $i = 1, \dots, N$ RVs ($N \geq n$), and $(\mathbf{x}_i, U_i)$ is independent of $(\mathbf{x}_j, U_j)$ for all $i \neq j$, where $U_i = Y_i - E[Y_i | \mathbf{x}_i]$.

Assumptions A1–A5 are ordered from least to most restrictive; A6(i) is the most restrictive but yields an exact CI.

Assumptions A1–A6 constitute a linear regression model:

$$Y_i = \mathbf{x}_i' \beta + U_i.$$

Remark. Everything so far is probability and statistics; there is no explicit connection to economics. To apply these techniques to economic phenomena, we must connect the probability model to economic theory. Key takeaway: except in a very special case (A1–A5 plus A6(i)), any CI is an approximation; the quality of approximation depends on how well finite-sample information approximates information in a hypothetical infinite sample.

3 Regression Modeling and M-Estimation

Recall the linear regression (LR) model:

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + U_i,$$

which (under assumptions A1–A6) provides a method to obtain inference on the conditional mean $E[Y | X]$, a feature of the conditional distribution $(Y | X)$.

A key question: why should we care about $E[Y | X]$?

Answer. Under the simple counting model, of course we can do some things, such as forecasting. But if we stop there, the conditional expectation we get from regression is essentially just a statistical object. (**Recall:** even the idea that $(Y | X)$ exists is an assumption we impose.)

Common side-questions / terminology: What do people mean by true distribution or true model? What do people mean by true parameters? What do people mean by mis-specification? What do people mean by large sample...

To endow $(Y | X)$ with economic meaning and to put the side-questions into a practically useful context, we need economic models. Useful economic models:

1. Characterise economic phenomena in a useful way:
 - They are rigorous quantitative representations.
 - They incorporate uncertainty through meaningful random variables.
 - They guarantee a unique solution for variables that can be meaningfully linked to observed data.
2. Often describe an economic equilibrium (micro/macro theory).
3. Can describe the outcome of a controlled experiment (experimental economics).

Economic models have the following elements:

- **Primitive shocks:** a set of $K_\varepsilon \times 1$ random vectors $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$, which are independent of each other and of any exogenous variables.
- **Exogenous variables:** a set of $K_{\tilde{\mathbf{x}}} \times 1$ vectors $\{\tilde{\mathbf{x}}_i\}_{i=1}^N$, treated as determined outside the model.
- **Structural parameters:** a $K_\theta \times 1$ vector $\boldsymbol{\theta}$, whose elements represent meaningful quantities within the model.
- **Endogenous variables:** a set of $K_w \times 1$ vectors $\{\mathbf{w}_i\}_{i=1}^N$, uniquely determined by the model in terms of primitive shocks, exogenous variables, and structural parameters.

Econometrician: the matchmaker that “ships” a statistical model (e.g. conditional mean regression) with (a class of) economic models.

- Stronger link $\Rightarrow$ more reliable economics conclusion.
- Weaker link $\Rightarrow$ less reliable economic conclusion.
- No link $\Rightarrow E[Y | \tilde{\mathbf{x}}]$ is a purely statistical quantity (e.g. best predictor).

3.1 Economic Models

A useful micro- or macro-economic model, indexed by m , characterises an equilibrium through a system of equations:

$$\mathcal{F}_m\left(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \{\boldsymbol{\varepsilon}_i\}, \boldsymbol{\theta}\right) = \mathbf{0},$$

which guarantees a unique solution:

$$\mathbf{w}_i = g_{mi}\left(\{\tilde{\mathbf{x}}_i\}, \{\boldsymbol{\varepsilon}_i\}, \boldsymbol{\theta}\right), \quad i = 1, \dots, n,$$

although it is typically not possible to obtain $g_{mi}(\cdot)$ explicitly.

In general, $\mathbf{w}_i$, $\tilde{\mathbf{x}}_i$, $\boldsymbol{\varepsilon}_i$, and $\boldsymbol{\theta}$ are specific to a given model m (and so are $K_w, K_{\tilde{x}}, K_{\varepsilon}, K_{\theta}$).

If we consider a class of economic models $m = 1, \dots, M$, it is often possible to identify common components:

- $\mathbf{w}_i = (\mathbf{w}'_{i,0}, \mathbf{w}'_{i,m})'$, where $\mathbf{w}_{i,0}$ are common to all models, and $\mathbf{w}_{i,m}$ are specific to model m .
- $\tilde{\mathbf{x}}_i = (\tilde{\mathbf{x}}'_{i,0}, \tilde{\mathbf{x}}'_{i,m})'$, where $\tilde{\mathbf{x}}_{i,0}$ are common to all models, and $\tilde{\mathbf{x}}_{i,m}$ are specific to model m .
- $\boldsymbol{\varepsilon}_i = (\boldsymbol{\varepsilon}'_{i,0}, \boldsymbol{\varepsilon}'_{i,m})'$, where $\boldsymbol{\varepsilon}_{i,0}$ have the same meaning across all models, and $\boldsymbol{\varepsilon}_{i,m}$ are specific to model m .
- $\boldsymbol{\theta} = (\boldsymbol{\theta}'_0, \boldsymbol{\theta}'_m)'$, where $\boldsymbol{\theta}_0$ are parameters with common meaning across all models, and $\boldsymbol{\theta}_m$ are specific to model m .

Example 3.1.1. Suppose w_{i1} (the first element of $\mathbf{w}_i$) and $\tilde{x}_{i1}$ (the first element of $\tilde{\mathbf{x}}_i$) are common to all models. We may define $\boldsymbol{\theta}$ such that

$$\frac{dw_{i1}}{d\tilde{x}_{i1}} = \theta_1, \quad \theta_1 \in \boldsymbol{\theta}_0.$$

This makes sense because $\frac{dw_{i1}}{d\tilde{x}_{i1}}$ quantifies a relationship between variables common to all models in the class.

3.2 The Matchmaking: LR and Economic Theory

We have data, assumed generated by random variables Y_i and $\mathbf{x}_i$, $i = 1, \dots, n$. We consider inference on $E[Y_i | \mathbf{x}_i]$ using the LR model

$$Y_i = \mathbf{x}'_i \boldsymbol{\beta} + U_i$$

and Assumptions A1–A5 plus either A6(i) or A6(ii).

We can match the LR to:

- **a specific model m :**
 - Obtain inference on model-specific effects,
 - Test model-specific implications,
 - Evaluate how well the economic model fits the data (relative to others).

- **a class of models** $m = 1, \dots, M$:
 - Obtain robust inference on effects with common meaning,
 - Develop “stylised facts” to help improve theoretical models.

Example 3.2.1 (Automotive supply and demand). Suppose we have data on quantities sold, prices, and characteristics of automobiles (non-price characteristics include size, power, fuel consumption, etc.). We want inference on demand elasticities (own- and cross- prices and characteristics). These elasticities quantify substitution patterns and are useful for policy.

If we formulate a LR, key practical questions are:

1. How do we assign data to Y_i and $\mathbf{x}_i$?
2. How do we interpret and verify assumptions on U_i ?
3. How do we interpret β ?

3.3 BLP (1995, Econometrica): Differentiated Products

BLP develops a supply-demand model for markets with differentiated products (e.g. automobiles). There are H households and N products. Each household h has income y_h and preferences for product characteristics

$$\phi_h = (\phi_{h1}, \dots, \phi_{hK})'.$$

Utility of household h for product i is

$$u_{ih} = h(y_h - p_i) \exp \left\{ \tilde{\mathbf{x}}_i' \phi_h + \xi_i + \zeta_{ih} \right\},$$

where

- p_i is the price of product i ;
- $\tilde{\mathbf{x}}_i$ are K characteristics of product i for which households have heterogeneous preferences;
- ξ_i is the aggregate effect of product characteristics for which households have homogeneous preferences;
- ζ_{ih} is the aggregate effect of idiosyncratic tastes, i.i.d. across households and products;
- $h(\cdot)$ is a strictly positive function.

BLP make the following general assumptions:

- ζ_{ih} follows the extreme value distribution.
- (y_h, ϕ_h) are distributed across households independent of $\{\tilde{\mathbf{x}}_i\}$, with density $f_{y,\phi}$, $E[\phi_h | y_h] \equiv \mu_\phi$ and $V(\phi_h | y_h) \equiv V_\phi$.
- Each household chooses the one product that yields maximum utility u_{ih} from the set of N products.

BLP also consider specific models of utility and price-setting:

- $m = 1$: $h(y_h - p_i) = (y_h - p_i)^{\alpha_1}$ and firms are perfectly competitive.

- $m = 2$: $h(y_h - p_i) = \exp\{-\alpha_2 p_i\}$, $\mathbf{V}_\phi = \mathbf{0}$, and firms are perfectly competitive.
- $m = 3$: $h(y_h - p_i) = (y_h - p_i)^{\alpha_1}$ and firms are monopolistic.
- $m = 4$: $h(y_h - p_i) = \exp\{-\alpha_2 p_i\}$, $\mathbf{V}_\phi = \mathbf{0}$, and firms are monopolistic.

In the BLP setting, the relative market shares are given by:

$$\ln(s_i/s_0) = \mathcal{D}_{mi}\left(\{\ln p_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)'\right), \quad i = 1, \dots, N,$$

where $s_0 = 1 - \sum_{i=1}^N s_i$ and $\alpha_3 = \alpha_1$, $\alpha_4 = \alpha_2$.

Prices are determined by the type of competition assumed:

- **Perfect competition:** prices equal marginal costs:

$$\ln p_i = \tilde{\mathbf{x}}_i' \boldsymbol{\gamma} + \omega_i, \quad i = 1, \dots, N.$$

- **Monopolistic competition:** prices have endogenous mark-ups:

$$\ln p_i = \mathcal{S}_{mi}\left(\{\ln s_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, \{\omega_j\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)'\right), \quad i = 1, \dots, N.$$

Set

$$\mathbf{w}_i = (\ln(s_i/s_0), \ln p_i)', \quad \boldsymbol{\varepsilon}_i = (\xi_i, \omega_i)', \quad \boldsymbol{\theta} = (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)', \quad \boldsymbol{\theta}_0 = (\boldsymbol{\phi}', \boldsymbol{\gamma}')'.$$

For $m = 1$ and $m = 2$, equilibrium is characterised by

$$\mathcal{F}_m(\cdot) \equiv \begin{pmatrix} \ln(s_1/s_0) - \mathcal{D}_{m1}(\{\ln p_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \vdots \\ \ln(s_N/s_0) - \mathcal{D}_{mN}(\{\ln p_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \ln p_1 - \tilde{\mathbf{x}}_1' \boldsymbol{\gamma} - \omega_1 \\ \vdots \\ \ln p_N - \tilde{\mathbf{x}}_N' \boldsymbol{\gamma} - \omega_N \end{pmatrix} = \mathbf{0}.$$

For $m = 3$ and $m = 4$, equilibrium is characterised by

$$\mathcal{F}_m(\cdot) \equiv \begin{pmatrix} \ln(s_1/s_0) - \mathcal{D}_{m1}(\{\ln p_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \vdots \\ \ln(s_N/s_0) - \mathcal{D}_{mN}(\{\ln p_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \ln p_1 - \mathcal{S}_{m1}(\{\ln s_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, \{\omega_j\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)') \\ \vdots \\ \ln p_N - \mathcal{S}_{mN}(\{\ln s_j\}, \{\tilde{\mathbf{x}}_j\}, \{\xi_j\}, \{\omega_j\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)') \end{pmatrix} = \mathbf{0}.$$

Since we are mainly interested in **demand elasticities**, focus on the demand equations and consider a linear approximation to $\mathcal{D}_m(\cdot)$.

If $\mathcal{D}_{mi}(\cdot)$ is differentiable w.r.t. $\ln p_i$, $\tilde{\mathbf{x}}_i$, and ξ_i , a first-order Taylor expansion around

$$p_i \approx p^*, \quad \tilde{\mathbf{x}}_i \approx \mathbf{0}, \quad \xi_i \approx 0$$

yields

$$\ln(s_i/s_0) \approx \delta_0 + \delta_1 \ln p_i + \delta_2 \sum_{j \neq i} \ln p_j + \tilde{\mathbf{x}}_i' \boldsymbol{\delta}_3 + \sum_{j \neq i} \tilde{\mathbf{x}}_j' \boldsymbol{\delta}_4 + U_i,$$

where

$$\delta_1 = \frac{\partial \ln(s_i/s_0)}{\partial \ln p_i}, \quad \delta_2 = \frac{\partial \ln(s_i/s_0)}{\partial \ln p_j}, \quad \delta_3 = \frac{\partial \ln(s_i/s_0)}{\partial \tilde{\mathbf{x}}_i}, \quad \delta_4 = \frac{\partial \ln(s_i/s_0)}{\partial \tilde{\mathbf{x}}_j},$$

evaluated at $p_i = p^*$, $\tilde{\mathbf{x}}_i = \mathbf{0}$, $\xi_i = 0$, and

$$U_i = \xi_i + \rho \bar{\xi}, \quad \bar{\xi} = \frac{1}{N} \sum_{i=1}^N \xi_i.$$

The linear approximation is more accurate when p_i , $\tilde{\mathbf{x}}_i$, and ξ_i are less dispersed and close to p^* , $\mathbf{0}$, 0.

3.4 The Match Between LR and Linearised BLP

We can establish a match as follows:

- $n = N$, $k = 1 + 2(1 + K_{\tilde{\mathbf{x}}})$.
- $Y_i = \ln(s_i/s_0)$,

$$\mathbf{x}_i = \left(1, \ln p_i, \sum_{j \neq i} \ln p_j, \tilde{\mathbf{x}}'_i, \sum_{j \neq i} \tilde{\mathbf{x}}'_j\right)'.$$

- $\boldsymbol{\beta} = \boldsymbol{\delta} \equiv (\delta_0, \delta_1, \delta_2, \boldsymbol{\delta}'_3, \boldsymbol{\delta}'_4)'$.
- $U_i = \xi_i + \rho \bar{\xi}$.

Note that $\boldsymbol{\delta}$ contains constant demand elasticities, evaluated at $p_i \approx p^*$, $\tilde{\mathbf{x}}_i \approx \mathbf{0}$, $\xi_i \approx 0$. We match these constant elasticities to moments of $(Y_i, \mathbf{x}_i)$: $E[\mathbf{x}_i Y_i]$ and $E[\mathbf{x}_i \mathbf{x}'_i]$.

- One advantage: these constant elasticities are robust to specifications of $f_{y,\phi}$.
- One disadvantage: we cannot measure how elasticities vary with prices, market shares, and characteristics.

If the LR satisfies $E[U_i | \mathbf{x}_i] = 0$ (A1), then we can only match it to a BLP model where

$$E[\xi | \{p_i\}, \{\tilde{\mathbf{x}}_i\}] = 0.$$

Suppose

$$E[\xi_i | \{\tilde{\mathbf{x}}_i\}] = 0, \quad E[\omega_i | \{\tilde{\mathbf{x}}_i\}] = 0, \quad E[\xi_i \omega_i | \{\tilde{\mathbf{x}}_i\}] = 0.$$

Then for $m = 1$ and $m = 2$, p_i is essentially exogenous. In this case, an LR satisfying $E[U_i | \mathbf{x}_i] = 0$ can be matched to $m = 1$ or $m = 2$, but **not** $m = 3$ or $m = 4$.

For $m = 1$ or $m = 2$, suppose $C(\xi_i, \bar{\xi} | \{p_i\}, \{\tilde{\mathbf{x}}_i\}) = 0$ and

$$V(\xi_i | \{p_i\}, \{\tilde{\mathbf{x}}_i\}) = V(\xi_i) = \sigma_\xi^2.$$

Hence,

$$C(U_i, U_j | X) = C(U_i, U_j) = \rho \sigma_\xi^2 / N.$$

If the LR satisfies $C(U_i, U_j | X) = 0$ (A4), then we can only match it to a BLP model with $\rho = 0$. This is satisfied by $m = 2$ (in which case also $\delta_2 = 0$ and $\boldsymbol{\delta}_4 = \mathbf{0}$) but not $m = 1$.

In summary:

- A LR satisfying A1–A5 and either A6(i) or A6(ii) can only be matched to a differentiated product market model with no interaction between product/consumer characteristics and with exogenous prices.

BLP have a lot to say about implications on substitution patterns of the utility specification in $m = 2$ (and $m = 4$).

- To derive a statistical model that matches to $m = 1$, we can start with the LR, but relax A4–A5. Implications:
 - The OLS estimator $\hat{\beta}$ is still unbiased but not efficient anymore.
 - A6(i) is not useful in constructing CIs.
 - A6(ii) is not satisfied for any finite N because $C(U_i, U_j | X) = C(U_i, U_j) = \rho\sigma_\xi^2/N$, but the dependence vanishes as $N \rightarrow \infty$.
 - We can replace A6(ii) with an alternative assumption such that the (hypothetical) OLS estimator $\hat{\beta}$ is still asymptotically normal.
 - A (hypothetical) LR with $N \rightarrow \infty$ matches to a BLP model with “infinitely many products”, each satisfying identical assumptions.
- To derive a statistical model that matches to $m = 3$ or $m = 4$, we need a simultaneous equations model (Week 6).

3.5 Robustifying the LR Match

What about matching to BLP models with

$$V(\xi_i | \{p_i\}, \{\tilde{\mathbf{x}}_i\}) = \sigma_{\xi_i}^2 ?$$

If ξ_i and ξ_j remain conditionally uncorrelated, such models can be matched to a (hypothetical) LR with heteroskedastic residuals:

$$E(U_i^2 | X) = \sigma_{u_i}^2.$$

Extending the LR to heteroskedastic residuals is easy: we may use the same estimator $\hat{\beta}$ to construct CIs, but need to adjust the standard errors.

The adjusted SEs are obtained from a heteroskedasticity-robust estimator of the variance matrix $V(\hat{\beta} | X)$:

$$\hat{V}^{HR}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \hat{U}_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbf{X}'\mathbf{X})^{-1}.$$

- The robust estimator is widely used as the default variance matrix estimate in most applied work.
- In R, the `lm_robust` function in the `estimatr` package implements OLS with heteroskedasticity-robust SEs.

If we want inference on elasticities for products with different market shares, characteristics, and/or prices, the earlier linearisation (w.r.t. $\ln p_i$, $\tilde{\mathbf{x}}_i$, and ξ_i simultaneously) is not sufficiently flexible.

Alternative: linearise $\mathcal{D}_{mi}(\cdot)$ w.r.t. ξ_i only. This yields

$$\ln(s_i/s_0) \approx \mu_i(\{p_j\}, \{\tilde{\mathbf{x}}_j\}, \boldsymbol{\theta}) + \xi_i + \frac{1}{N} \sum_{j=1}^N \rho_i(\{p_j\}, \{\tilde{\mathbf{x}}_j\}, \boldsymbol{\theta}) \xi_j.$$

With exogenous prices, it can be matched to a non-linear statistical model:

$$Y_i = \mu_i(X, \boldsymbol{\vartheta}) + U_i,$$

where

$$C(U_i, U_j | X) = \sigma_\xi^2 \rho_i(\{p_j\}, \{\tilde{\mathbf{x}}_j\}, \boldsymbol{\theta}) / N,$$

and

$$V(U_i | X) = \sigma_\xi^2 \left(1 + \frac{2\rho_i(\cdot)}{N} + \sum_{j=1}^N \left(\frac{\rho_i(\cdot)}{N} \right)^2 \right) \equiv \sigma_{u_i}^2(X, \boldsymbol{\vartheta}).$$

Note again that $C(U_i, U_j | X) \rightarrow 0$ and $V(U_i | X) \rightarrow \sigma_\xi^2$ as $N \rightarrow \infty$.

3.6 M-Estimation

There are many non-linear statistical models that allow for more flexibility at the expense of more complexity / less parsimony: Non-linear Least Squares, Quantile Regressions, Semi-Parametric models, etc.

Most estimators can be formulated as solutions to an optimisation problem of the form:

$$\hat{\boldsymbol{\vartheta}} = \arg \min_{\boldsymbol{\vartheta}} S_n(\boldsymbol{\vartheta}), \quad S_n(\boldsymbol{\vartheta}) = \frac{1}{n} \sum_{i=1}^n s(Y_i, \mathbf{x}_i, \boldsymbol{\vartheta}).$$

These are often referred to as M-Estimators. Note that $S_n(\boldsymbol{\vartheta})$ is an objective function that can be evaluated using available data and a particular value of $\boldsymbol{\vartheta}$, given $s(\cdot)$.

Example 3.6.1. OLS: $s(Y_i, \mathbf{x}_i, \mathbf{b}) = (Y_i - \mathbf{x}_i' \mathbf{b})^2$.

For $i = 1, \dots, N$, $N \geq n$: Assume there exists a unique

$$\boldsymbol{\vartheta}_0 = \arg \min_{\boldsymbol{\vartheta}} E[S_N(\boldsymbol{\vartheta})].$$

Let

$$\tilde{\boldsymbol{\vartheta}} = \arg \min_{\boldsymbol{\vartheta}} S_n(\boldsymbol{\vartheta}).$$

Under assumptions in Hansen (2022, Ch. 22),

$$\sqrt{N}(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_\vartheta).$$

For the LR,

$$\boldsymbol{\beta} = \arg \min_{\mathbf{b}} E[(Y_i - \mathbf{x}_i' \mathbf{b})^2 | X]$$

We can adjust the LR, or another statistical model by re-formulating $S_N(\cdot)$ and checking that all technical assumptions are satisfied to find a corresponding M-estimator.

3.7 Summary

To obtain **economic inference** from data, we need to match a **statistical model** to a class of **economic models**.

The LR is a semi-parametric model that is practically simple to implement and has well-known properties. However, the LR model can often only be matched to a restrictive class of economic models, which limits the economic interpretation of the statistical results.

To obtain more robust inference, we may adjust the standard LR model to establish a match to a more robust class of economic models. One simple adjustment is to focus on quantiles instead of the conditional mean, which leads to the QR (and the QR can be further adjusted).

In practice, it helps to view the OLS and QR estimators as special cases of a more general methodology: M-estimation. Because we know the general properties, this helps us deduce practical inference methods for the adjusted statistical models.

4 The Likelihood Function and Maximum Likelihood Estimation

4.1 The Maximum Likelihood Principle

Suppose we have a realisation $\{y_i, x_{i1}, \dots, x_{ik}\}_{i=1}^n$ of the random variables $\{Y_i, X_{i1}, \dots, X_{ik}\}_{i=1}^n$ and we are considering a family of distribution models for $(\mathbf{y}, \mathbf{X})$, denoted by F .

Definition 4.1.1 (Maximum Likelihood Principle (MLP)). Choose the distribution model within F that maximises the likelihood of randomly drawing the given realisation.

Formally: choose the distribution whose PDF evaluated at the realisation,

$$f(y_1, \dots, y_n, x_{11}, \dots, x_{nk}),$$

is the maximum in F .

If two or more PDFs attain the maximum value, then the distributions are called observationally equivalent.

Remark. This is a purely statistical principle. All of the parameters here are just statistical parameters.

Example 4.1.2. Suppose $n = 10$ observations are

$$5, 0, 1, 1, 0, 3, 2, 3, 4, 1.$$

If F contains all possible distributions, the MLP dictates that we should trivially choose

$$\Pr(Y_1 = 5, Y_2 = 0, Y_3 = 1, Y_4 = 1, Y_5 = 0, Y_6 = 3, Y_7 = 2, Y_8 = 3, Y_9 = 4, Y_{10} = 1) = 1.$$

However, F is not very useful for extracting useful information from this data.

Key question: what are we looking for? MLE is goal specific! Suppose the goal is to predict Y_{11} .

Then, a useful family may consist of independent and identical Poisson distributions with unknown mean θ , which we denote by F_θ .

Within F_θ , each candidate distribution is characterised by

$$f(y_1, \dots, y_n; \theta) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{y_i}}{y_i!}, \quad \text{for any } n = 1, \dots, 11.$$

As in Lecture 3, we again refer to θ as a (statistical) parameter, which is not a random variable, and in this case, F_θ as a parametric family.

For the given realisation,

$$f(y_1, \dots, y_{10}; \theta) = \frac{e^{-10\theta} \theta^{\sum_{i=1}^{10} y_i}}{\prod_{i=1}^{10} y_i!} = \frac{e^{-10\theta} \theta^{20}}{207,365}.$$

The MLP now boils down to: choose θ to maximise the parametric PDF $f(y_1, \dots, y_{10}; \theta)$.

Definition 4.1.3. The parametric PDF is called the likelihood function of θ , denoted $L(\theta; D)$, where $D = \{y_1, \dots, y_{10}\}$ is the set of all realisations.

Because maximising $L(\theta; D)$ subject to θ is equivalent to maximising any monotonic transformation of $L(\theta; D)$, it is often convenient to work with the log-likelihood $\ln L(\theta; D)$.

In this example, we choose

$$\theta^* = \arg \max_{\theta} \ln L(\theta; D) = \arg \max_{\theta} (-10\theta + 20 \ln \theta - 207,365) = 2.$$

Hence, we would forecast Y_{11} from a Poisson distribution with PDF

$$f(y_{11}) = \frac{e^{-2} 2^{y_{11}}}{y_{11}!}.$$

Note that $f(y_{11})$ characterises all forecast uncertainty: we can derive the expectation, variance, quantiles, etc. It is **Not** a point estimate.

4.2 The MLP in Econometrics

Recall that the equilibrium of an economic model with a single endogenous variable Y_i can be expressed

$$Y_i = g_{mi}(\mathbf{x}_i, \{\epsilon_i\}, \boldsymbol{\theta}), \quad i = 1, \dots, n,$$

where the vector $\mathbf{x}_i$ contains strictly exogenous variables, and $\{\epsilon_i\}$ are primitive shocks, distributed independent of $\{\mathbf{x}_i\}$.

The PDF of the joint distribution $(\{\mathbf{x}_i\}, \{\epsilon_i\})$, if one exists, is

$$\begin{aligned} f_m(\{\mathbf{x}_i\}, \{\epsilon_i\}; \boldsymbol{\theta}) &= f_m(\{\mathbf{x}_i\}; \boldsymbol{\theta}) \prod_{i=1}^n f_m(\epsilon_i; \boldsymbol{\theta}), \quad (\text{independence}) \\ &= f(\{\mathbf{x}_i\}) \prod_{i=1}^n f_m(\epsilon_i; \boldsymbol{\theta}). \quad (\text{exogeneity}) \end{aligned}$$

Hence, specifying $f_m(\epsilon_i; \boldsymbol{\theta})$, for $i = 1, \dots, n$, through

$$f_m(\mathbf{X}, \boldsymbol{\epsilon}; \boldsymbol{\theta}) \xrightarrow{\text{through } g_{mi}} f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\theta}),$$

implies the PDF

$$f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\theta}) = f_m(\mathbf{X}; \boldsymbol{\theta}) f_m(\mathbf{y} \mid \mathbf{X}; \boldsymbol{\theta}) = f(\mathbf{X}) f_m(\mathbf{y} \mid \mathbf{X}; \boldsymbol{\theta}).$$

Since ϵ_i are primitive shocks in model m , the PDFs $f_m(\epsilon_i; \boldsymbol{\theta})$ and $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\theta})$ are specific to a given economic model.

Note that we have written the PDFs $f_m(\epsilon_i; \boldsymbol{\theta})$ and $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\theta})$ as functions of random variables, which are defined within a particular economic model.

To obtain a likelihood, we must assume that we have a realisation

$$D = \{y_1, \dots, y_n, x_{11}, \dots, x_{nk}\}$$

from RVs that match precisely to the random vectors $\mathbf{y}$ and $\mathbf{X}$ defined in the economic model. Given the match, we may view the PDF $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\theta})$ evaluated at D as the likelihood $L(\boldsymbol{\theta}; D)$ for the structural parameters in model m .

Suppose we consider a family $F_{\boldsymbol{\vartheta}}$ of statistical models for

$$(Y_1, \dots, Y_n, X_{11}, \dots, X_{nk})$$

that match precisely to the economic model m .

- There exists some $\boldsymbol{\vartheta}_0 = \boldsymbol{\theta}$
- MLP: choose the distribution with PDF $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}^*)$.
- $\boldsymbol{\vartheta}^*$ depends on $f_m(\epsilon_i; \boldsymbol{\theta})$ and $g_{mi}(\cdot)$, but not $f(\mathbf{X})$.

Because $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\theta}) = f(\mathbf{X}) f_m(\mathbf{y} \mid \mathbf{X}; \boldsymbol{\theta})$ and $f(\mathbf{X})$ does not depend on $\boldsymbol{\vartheta}$,

$$\boldsymbol{\vartheta}^* = \arg \max_{\boldsymbol{\vartheta}} f_m(y_1, \dots, y_n, x_{11}, \dots, x_{nk}; \boldsymbol{\vartheta}) = \arg \max_{\boldsymbol{\vartheta}} f_m(y_1, \dots, y_n \mid x_{11}, \dots, x_{nk}; \boldsymbol{\vartheta}).$$

What if we consider matching statistical models such as the linear regression (LR) to a class of economic models?

Matching the LR to BLP models:

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + U_i, \quad i = 1, \dots, n,$$

where

- $\boldsymbol{\beta}$ (some moments) is a function of $\boldsymbol{\theta}$ (parameter with economics meaning);
- $\mathbf{x}_i$ is function of $\{p_i\}$ and $\{\tilde{\mathbf{x}}_i\}$;
- $U_i = \xi_i + \rho \bar{\xi}$.

Recall that $\{\xi_i\}$ are (primitive) household utility shocks: the distribution of $\{\xi_i\}$ has PDF

$$f(\{\xi_i\}; \boldsymbol{\theta}) = \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta}).$$

Because $\{\xi_i\}$, $\{\tilde{\mathbf{x}}_i\}$ are independent,

$$f(\{\tilde{\mathbf{x}}_i\}, \{\xi_i\}; \boldsymbol{\theta}) = f(\{\tilde{\mathbf{x}}_i\}) \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta}).$$

If firms are perfectly competitive, $\{p_i\}$ and $\{\xi_i\}$ are independent, such that

$$f(\{p_i\}, \{\tilde{\mathbf{x}}_i\}, \{\xi_i\}; \boldsymbol{\theta}) = f(\{p_i\}, \{\tilde{\mathbf{x}}_i\}) \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta}) = f(\mathbf{X}) \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta}),$$

and therefore, using conditional probability,

$$f(\{\xi_i\} \mid \mathbf{X}; \boldsymbol{\theta}) = \frac{f(\mathbf{X}, \{\xi_i\}; \boldsymbol{\theta})}{f(\mathbf{X})} = \frac{f(\mathbf{X}) \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta})}{f(\mathbf{X})} = \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta}).$$

We use $f(\{\xi_i\} \mid \mathbf{X}; \boldsymbol{\theta}) = \prod_{i=1}^n f(\xi_i; \boldsymbol{\theta})$ and $U_i = \xi_i + \rho \bar{\xi}$ to derive the PDF for $(\mathbf{u} \mid \mathbf{X})$, and consequently, $(\mathbf{y} \mid \mathbf{X})$.

BLP, like most economic models, does not explicitly detail $f(\xi_i; \boldsymbol{\theta})$: any additional specification of $f(\xi_i; \boldsymbol{\theta})$ restricts the class of BLP models.

Key Question: in what sense is this important? How does one specification of $f(\xi_i; \boldsymbol{\theta})$ vs another specification affect the model implications?

Start with $\xi_i \sim \mathcal{N}(0, \sigma_{\xi_i}^2)$: symmetric at zero, mesokurtic, defined by $\sigma_{\xi_i}^2$. If any property of the Normal distribution is restrictive, how and why?

- Perhaps not useful to model ξ_i as a primitive shock?
- If yes, then model it in terms of more primitive components!

If a match to BLP models with different $f(\xi_i; \boldsymbol{\theta})$ specifications is desired, then derive and analyse all implied likelihoods!

Example 4.2.1 (Normal shocks and $\rho = 0$). Simplest example: $\xi_i \sim \mathcal{N}(0, \sigma_{\xi}^2)$ and $\rho = 0$. Then, $(U_i \mid \mathbf{X}) \sim \mathcal{N}(0, \sigma_{\xi}^2)$ and

$$f(U_i \mid \mathbf{x}_i; \sigma_{\xi}^2) = \frac{1}{\sqrt{2\pi\sigma_{\xi}^2}} \exp\left\{-\frac{U_i^2}{2\sigma_{\xi}^2}\right\},$$

which implies

$$f(Y_i \mid \mathbf{x}_i; \sigma_{\xi}^2) = \frac{1}{\sqrt{2\pi\sigma_{\xi}^2}} \exp\left\{-\frac{(Y_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{2\sigma_{\xi}^2}\right\}.$$

For a given realisation $D = \{y_1, \dots, y_n, x_{11}, \dots, x_{nk}\}$, the likelihood is

$$L(\boldsymbol{\beta}, \sigma_{\xi}^2; D) = \frac{f(x_{11}, \dots, x_{nk})}{(2\pi\sigma_{\xi}^2)^{n/2}} \exp\left\{-\frac{1}{2\sigma_{\xi}^2} \sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_{ij}\beta_j\right)^2\right\},$$

and the log-likelihood is

$$\ln L(\boldsymbol{\beta}, \sigma_{\xi}^2; D) = \ln f(x_{11}, \dots, x_{nk}) - \frac{n}{2} \ln(2\pi\sigma_{\xi}^2) - \frac{1}{2\sigma_{\xi}^2} \sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_{ij}\beta_j\right)^2.$$

Maximising $\ln L(\boldsymbol{\beta}, \sigma_\xi^2; D)$ yields

$$\boldsymbol{\beta}^* = \left(\sum_{i=1}^n \begin{pmatrix} x_{i1}^2 & \cdots & x_{i1}x_{ik} \\ \vdots & \ddots & \vdots \\ x_{ik}x_{i1} & \cdots & x_{ik}^2 \end{pmatrix} \right)^{-1} \sum_{i=1}^n \begin{pmatrix} x_{i1} \\ \vdots \\ x_{ik} \end{pmatrix} y_i, \quad \sigma_\xi^{*2} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_{ij} \beta_j^* \right)^2.$$

Compare these to:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad \hat{\sigma}_y^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

$\boldsymbol{\beta}^*, \sigma_\xi^{*2}$ are equivalent to a realisation of $\hat{\boldsymbol{\beta}}, \hat{\sigma}_y^2$.

4.3 Maximum Likelihood Estimator

Why is the MLP useful? It underpins the statistical method: maximum likelihood estimation.

Definition 4.3.1. Suppose we consider the PDF $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})$ as a function of random variables, and set

$$\hat{\boldsymbol{\vartheta}} = \arg \max_{\boldsymbol{\vartheta}} f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}).$$

$\hat{\boldsymbol{\vartheta}}$ is a random variable since it is a function of $\mathbf{y}$ and $\mathbf{X}$; we call it the Maximum Likelihood Estimator (MLE).

$$\boldsymbol{\vartheta}^* = \arg \max_{\boldsymbol{\vartheta}} L(\boldsymbol{\vartheta}; D)$$

is one realisation of the MLE, obtained by evaluating $\hat{\boldsymbol{\vartheta}}$ at D , which is one realisation of $\mathbf{y}$ and $\mathbf{X}$.

The key concept underlying maximum likelihood estimation is the following assumption.

Assumption 4.3.2. Within $F_{\boldsymbol{\vartheta}}$, there exists an “ideal” distribution with PDF $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$, such that for every $\boldsymbol{\vartheta}$,

$$\mathbb{E}[\ln f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})] = \int_{(\mathbf{y}, \mathbf{X})} \ln f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}) f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0) d(\mathbf{y}, \mathbf{X}) < \infty.$$

Remark. The PDF $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$ is “ideal” in the sense that it can be matched to a class of economic models: e.g., $\boldsymbol{\vartheta}_0$ matches to the structural parameters $\boldsymbol{\theta}$ in the BLP models.

Example 4.3.3. $\boldsymbol{\vartheta}_0$ is a function of $\boldsymbol{\vartheta}$ — e.g., $\boldsymbol{\beta}, \sigma_y^2$ in the LR are functions of structural BLP parameters.

The PDF $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$ satisfies a version of the MLP extended to the expected log-likelihood:

$$\boldsymbol{\vartheta}_0 = \arg \max_{\boldsymbol{\vartheta}} \mathbb{E}[\ln f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})],$$

because for every $\boldsymbol{\vartheta} \neq \boldsymbol{\vartheta}_0$, Jensen's Inequality yields:

$$\begin{aligned}\mathbb{E}[\ln f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})] - \mathbb{E}[\ln f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)] &= \mathbb{E}\left[\ln \left(\frac{f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})}{f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)} \right)\right] \\ &\leq \ln \left(\mathbb{E} \left[\frac{f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})}{f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)} \right] \right) = 0.\end{aligned}$$

Remark. Many textbooks refer to $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$ as the PDF of a true distribution. In economics, the concept of a true distribution is not necessary, and often not useful.

4.3.1 Specification Uncertainty

Specification uncertainty is inevitable. Suppose we consider M families $F_{1, \boldsymbol{\vartheta}_1}, \dots, F_{M, \boldsymbol{\vartheta}_M}$ as reasonable candidates to model $(\mathbf{y}, \mathbf{X})$. Setting

$$\boldsymbol{\vartheta}_0 = (\boldsymbol{\vartheta}'_1, \dots, \boldsymbol{\vartheta}'_M)',$$

an “ideal” PDF may be formulated:

$$f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0) = \omega_1 f_1(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_1) + \dots + \omega_M f_M(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_M),$$

with weights $\omega_1, \dots, \omega_M$ non-negative and $\sum_{m=1}^M \omega_m = 1$.

Note that if $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_m) = f(\mathbf{X})f_m(\mathbf{y} | \mathbf{X}; \boldsymbol{\vartheta}_m)$, then

$$f(\mathbf{y} | \mathbf{X}; \boldsymbol{\vartheta}_0) = \omega_1 f_1(\mathbf{y} | \mathbf{X}; \boldsymbol{\vartheta}_1) + \dots + \omega_M f_M(\mathbf{y} | \mathbf{X}; \boldsymbol{\vartheta}_M),$$

$$\mathbb{E}[g(\mathbf{y}) | \mathbf{X}] = \omega_1 \mathbb{E}_1[g(\mathbf{y}) | \mathbf{X}] + \dots + \omega_M \mathbb{E}_M[g(\mathbf{y}) | \mathbf{X}].$$

How do we set the weights? Decision Theory:

$$\{\hat{\omega}_m\} = \arg \max_{\omega_1, \dots, \omega_M} \sum_{m=1}^M \omega_m \ln \left(\omega_m f_m(\mathbf{y} | \mathbf{X}; \hat{\boldsymbol{\vartheta}}_m) \right),$$

where

$$\hat{\boldsymbol{\vartheta}}_m = \arg \max_{\boldsymbol{\vartheta}_m} f_m(\mathbf{y} | \mathbf{X}; \boldsymbol{\vartheta}_m).$$

Solution:

$$\hat{\omega}_m = \frac{f_m(\mathbf{y} | \mathbf{X}; \hat{\boldsymbol{\vartheta}}_m)}{\sum_{m=1}^M f_m(\mathbf{y} | \mathbf{X}; \hat{\boldsymbol{\vartheta}}_m)}.$$

Intuition: If $L_1(\boldsymbol{\vartheta}_1^*; D) > L_2(\boldsymbol{\vartheta}_2^*; D)$, then $F_{1, \boldsymbol{\vartheta}_1}$ is attributed more relative weight (relative to $F_{2, \boldsymbol{\vartheta}_2}$) in carrying out inference: $\{L_m(\boldsymbol{\vartheta}_m^*; D)\}$ quantify specification uncertainty.

Remark. (i) If you try many families and in the end their likelihood values are similar and the resulting intervals are also similar, that suggests specification uncertainty may not be very important, because your inference is very robust.

(ii) But if the likelihood values are very similar while the intervals, or the interpretations implied by the intervals, are completely different, then specification uncertainty is a real and important issue.

Alternative models can be compared using log-likelihoods or information criteria (IC). IC combine the log-likelihood with a function of k_m , which is the dimension of $\boldsymbol{\vartheta}_m$. General formula:

$$\text{IC}_m = -2 \ln L_m(\boldsymbol{\vartheta}_m^*; D) + c(k_m, n).$$

The term $c(k_m, n)$ is increasing in k_m and often interpreted as a penalty for including k_m unknowns in the model. IC_m is often interpreted as a tradeoff between flexibility (larger $L_m(\boldsymbol{\vartheta}_m^*; D)$) and parsimony (smaller k_m).

Popular information criteria:

- AIC: $c(k_m, n) = 2k_m$;
- BIC: $c(k_m, n) = k_m \ln n$;
- HQC: $c(k_m, n) = 2k_m \ln(\ln n)$.

Remark. (i) If several models have very similar likelihood or IC values, and they also produce similar intervals and conclusions, then your conclusion is fairly robust to specification, because whichever model you use gives almost the same inference.

- (ii) But if different models produce very different inference, then specification uncertainty is genuinely important and cannot be ignored.

4.3.2 Properties of the MLE

If $\ln f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}) = \sum_{i=1}^n \ln f(y_i, \mathbf{x}_i; \boldsymbol{\vartheta})$, then the MLE is also an M-Estimator with $s(y_i, \mathbf{x}_i; \boldsymbol{\vartheta}) = -\ln f(y_i, \mathbf{x}_i; \boldsymbol{\vartheta})$:

$$\hat{\boldsymbol{\vartheta}} = \arg \min_{\boldsymbol{\vartheta}} \frac{1}{n} \sum_{i=1}^n \left(-\ln f(y_i, \mathbf{x}_i; \boldsymbol{\vartheta}) \right).$$

Recall from Lecture 3 that for $i = 1, \dots, N$, $N \geq n$, the corresponding $\tilde{\boldsymbol{\vartheta}}$ satisfies

$$\sqrt{N}(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}_{\boldsymbol{\vartheta}}).$$

Setting

$$\mathbf{G}(\boldsymbol{\vartheta}) = \mathbb{E} \left[\frac{\partial s}{\partial \boldsymbol{\vartheta}} \frac{\partial s}{\partial \boldsymbol{\vartheta}'} \right], \quad \mathbf{H}(\boldsymbol{\vartheta}) = \mathbb{E} \left[\frac{\partial^2 s}{\partial \boldsymbol{\vartheta} \partial \boldsymbol{\vartheta}'} \right],$$

$\mathbf{V}_{\boldsymbol{\vartheta}}$ is given by

$$\mathbf{V}_{\boldsymbol{\vartheta}} = \mathbf{H}(\boldsymbol{\vartheta}_0)^{-1} \mathbf{G}(\boldsymbol{\vartheta}_0) \mathbf{H}(\boldsymbol{\vartheta}_0)^{-1}.$$

When $s(y_i, \mathbf{x}_i; \boldsymbol{\vartheta}) = -\ln f(y_i, \mathbf{x}_i; \boldsymbol{\vartheta})$, we have the information matrix equality:

$$\mathbf{G}(\boldsymbol{\vartheta}_0) = \mathbf{H}(\boldsymbol{\vartheta}_0).$$

Therefore, the MLE satisfies

$$\sqrt{N}(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{H}(\boldsymbol{\vartheta}_0)^{-1}).$$

- $N\mathbf{H}(\boldsymbol{\vartheta}_0)$ is called the Fisher Information Matrix.
- $\frac{1}{N}\mathbf{H}(\boldsymbol{\vartheta}_0)^{-1}$ is the asymptotic variance matrix of the MLE.
- $\frac{1}{N}\mathbf{H}(\boldsymbol{\vartheta}_0)^{-1}$ is equivalent to the Cramér-Rao Lower Bound: the MLE has the “smallest” possible asymptotic variance of all consistent estimators.
- This is a generalisation of the Gauss-Markov Theorem from Lecture 2.

4.3.3 Interpretation

The MLE is used to approximate an “ideal” PDF $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$, which is matched to (a class of) economic models.

- For inference from the matched economic models, the MLE provides the most efficient method for extracting useful information from the available data.

What if we aim to obtain inference from economic models that do not match precisely to a specific PDF from a parametric family $F_{\boldsymbol{\vartheta}}$? Question re-phrased: How do we deal with specification uncertainty?

4.3.4 Alternative Approaches to Specification Uncertainty

- **Option 1:** Consider M families $\{F_{m, \boldsymbol{\vartheta}_m}\}$ that match to alternative economic models and set the “ideal” PDF to

$$f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0) = \sum_{m=1}^M \omega_m f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_m).$$

- Use MLE to approximate each $f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_m)$, $m = 1, \dots, M$.
- Use $L_m(\boldsymbol{\vartheta}_m^*; D)$ to heuristically assess the relative weights $\{\hat{\omega}_m\}$.
- Obtain inference from

$$f(\mathbf{y}, \mathbf{X}; \hat{\boldsymbol{\vartheta}}) = \sum_{m=1}^M \hat{\omega}_m f_m(\mathbf{y}, \mathbf{X}; \hat{\boldsymbol{\vartheta}}_m)$$

or qualitatively interpret the results from $\{f_m(\mathbf{y}, \mathbf{X}; \hat{\boldsymbol{\vartheta}}_m)\}$ based on $\{\hat{\omega}_m\}$.

- **Option 2:** Consider M families $\{F_{m, \boldsymbol{\vartheta}_m}\}$ that match to alternative economic models and set the “ideal” PDF to

$$f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0) = \sum_{m=1}^M \omega_m f_m(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_m).$$

- Use Bayesian Methods for exact inference on $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$ (Lecture 5).
- **Option 3:** Treat the “ideal” PDF $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$ as unspecified, but $\boldsymbol{\vartheta}_0$ matches to some structural $\boldsymbol{\theta}$ in a class of economic models.
- Specify a function $s(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})$ that satisfies

$$\boldsymbol{\vartheta}_0 = \arg \min_{\boldsymbol{\vartheta}} \mathbb{E}[s(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})].$$

- Use another PDF $\tilde{f}(\cdot)$ and set $s(\cdot) = -\ln \tilde{f}(\cdot)$: Quasi-MLE with asymptotic variance

$$\frac{1}{N} \mathbf{H}(\boldsymbol{\vartheta}_0)^{-1} \mathbf{G}(\boldsymbol{\vartheta}_0) \mathbf{H}(\boldsymbol{\vartheta}_0)^{-1}.$$

- Use the Generalised Method of Moments (Lecture 6).

5 The Bayesian Framework

5.1 Econometric Inference and Sources of Information

Recall: For a distribution $(\mathbf{y}, \mathbf{X})$, the parametric PDF

$$f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta})$$

characterises a family of statistical models $\mathcal{F}_{\boldsymbol{\vartheta}}$ for extracting useful information from the data. The functional form of $f(\cdot)$ is defined by the family $\mathcal{F}_{\boldsymbol{\vartheta}}$, and the parameter vector $\boldsymbol{\vartheta}$ quantifies the unknown part of $\mathcal{F}_{\boldsymbol{\vartheta}}$.

To match the statistical models to economic models, assume there exists an “ideal” distribution in $\mathcal{F}_{\boldsymbol{\vartheta}}$ defined by $\boldsymbol{\vartheta}_0$.

Example 5.1.1 (Conditionally normal model).

$$f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0) = f(\mathbf{X}) \phi(\mathbf{y} \mid \mathbf{X}\boldsymbol{\beta}, \sigma_y^2 \mathbf{I}_n),$$

where $\phi(\mathbf{y} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the PDF of the multivariate normal distribution: $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Here,

$$\boldsymbol{\vartheta}_0 = (\boldsymbol{\beta}', \sigma_y^2).$$

Lecture 2 identities:

$$\boldsymbol{\beta} = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i Y_i], \quad \sigma_y^2 = \mathbb{E}[Y_i^2] - \mathbb{E}[\mathbf{x}_i Y_i]' \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i Y_i].$$

Essential statistical question: What can we learn from the data about the “ideal” distribution $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$?

Essential econometric question: What can we learn from the data about the economic models matched to $f(\mathbf{y}, \mathbf{X}; \boldsymbol{\vartheta}_0)$?

Learning from an estimator: use $\hat{\boldsymbol{\vartheta}}$ (e.g. MLE, OLS, etc.) to construct confidence intervals (CIs) for elements of $\boldsymbol{\vartheta}_0$.

Recall: except for OLS with A6(i: normality), CIs constructed from $\hat{\boldsymbol{\vartheta}}$ have no established properties with respect to $\boldsymbol{\vartheta}_0$.

CIs constructed from $\hat{\boldsymbol{\vartheta}}$ approximate CIs constructed from $\tilde{\boldsymbol{\vartheta}}$: the hypothetical estimator formulated using additional RVs

$$\{Y_{n+i}, \mathbf{x}_{n+i}\}_{i=1}^{\infty}$$

that satisfy additional assumptions.

CIs from $\tilde{\boldsymbol{\vartheta}}$ quantify uncertainty regarding statistical models in $\mathcal{F}_{\boldsymbol{\vartheta}}$ from which we interpret information in the data regarding economic phenomena.

Property 5.1.2 (Two sources of information). *Econometric inference always combines two sources of information:*

1. the econometrician provides information about the distribution $(\mathbf{y}, \mathbf{X})$;

2. the data provides information about the distribution $(\mathbf{y}, \mathbf{X})$.

MLE:

1. Econometrician provides information regarding the distribution $(\mathbf{y}, \mathbf{X})$, up to unknown $\boldsymbol{\vartheta}_0$.
2. Econometrician provides information regarding the distribution

$$(\{Y_{n+i}, \mathbf{x}_{n+i}\}_{i=1}^{\infty} \mid \mathbf{y}, \mathbf{X}),$$

up to unknown $\boldsymbol{\vartheta}_0$. (Independence, ...)

3. Data provides information regarding the unknown $\boldsymbol{\vartheta}_0$.

Less information from the econometrician means more flexibility for extracting information from data; more information from the econometrician means more parsimony in extracting information from data.

Efficiency = optimal balance between flexibility and parsimony.

5.2 Bayesian Framework

The Bayesian approach replaces the estimator with an alternative link between data and statistical models.

Assume there exists a random vector $\boldsymbol{\vartheta}$ such that the distribution $(\boldsymbol{\vartheta}, \mathbf{y}, \mathbf{X})$ has a PDF $f(\boldsymbol{\vartheta}, \mathbf{y}, \mathbf{X})$. The random vector $\boldsymbol{\vartheta}$ captures all available information about $\boldsymbol{\vartheta}_0$.

Definition 5.2.1 (Prior distribution). The marginal distribution with density

$$f(\boldsymbol{\vartheta}) = \int f(\boldsymbol{\vartheta}, \mathbf{y}, \mathbf{X}) d(\mathbf{y}, \mathbf{X})$$

is called the prior distribution and summarises all information about $\boldsymbol{\vartheta}_0$ available from sources outside the data.

Definition 5.2.2 (Posterior distribution). The conditional distribution $(\boldsymbol{\vartheta} \mid \mathbf{y}, \mathbf{X})$ with density $f(\boldsymbol{\vartheta} \mid \mathbf{y}, \mathbf{X})$, evaluated at the realisation D and denoted $(\boldsymbol{\vartheta} \mid D)$, is called the posterior distribution and summarises all information about $\boldsymbol{\vartheta}_0$ available from all sources, including the data.

The conditional density

$$f(\mathbf{y}, \mathbf{X} \mid \boldsymbol{\vartheta}) = \frac{f(\boldsymbol{\vartheta}, \mathbf{y}, \mathbf{X})}{f(\boldsymbol{\vartheta})}$$

is conceptually equivalent to the density $f(\mathbf{y}, \mathbf{X} \mid \boldsymbol{\vartheta}_0)$, with $\boldsymbol{\vartheta}_0$ replaced by $\boldsymbol{\vartheta}$. (Only changes from "indexing the distribution by a parameter" to "conditioning that distribution on a random variable.")

In practice, the most common approach is to specify:

1. the prior density $f(\boldsymbol{\vartheta})$ using information from economic theory, previous empirical work, etc.;
2. the conditional density $f(\mathbf{y}, \mathbf{X} \mid \boldsymbol{\vartheta})$ derived from $f(\mathbf{y}, \mathbf{X} \mid \boldsymbol{\vartheta}_0)$. (Recall lecture 4: both are derived from the economic model, the distribution of structural shocks, and equilibrium relationships)

Definition 5.2.3 (Bayesian likelihood). The density $f(\mathbf{y}, \mathbf{X} \mid \boldsymbol{\vartheta})$ evaluated at the realisation D is the Bayesian likelihood $f(D \mid \boldsymbol{\vartheta})$.

Definition 5.2.4 (Marginal likelihood). The marginal density

$$f(\mathbf{y}, \mathbf{X}) = \int f(\boldsymbol{\vartheta}, \mathbf{y}, \mathbf{X}) d\boldsymbol{\vartheta},$$

evaluated at the realisation D , is the marginal likelihood $f(D)$.

Theorem 5.2.5 (Bayes' theorem). Given the distribution $(\boldsymbol{\vartheta}, \mathbf{y}, \mathbf{X})$ with prior density $f(\boldsymbol{\vartheta})$ and likelihood $f(D \mid \boldsymbol{\vartheta})$, we have

$$\Pr(0 < f(D) < \infty) = 1,$$

and the posterior density is

$$f(\boldsymbol{\vartheta} \mid D) = \frac{f(\boldsymbol{\vartheta}) f(D \mid \boldsymbol{\vartheta})}{f(D)}.$$

This “result” is almost trivial!

The posterior density $f(\boldsymbol{\vartheta} \mid D)$ is a complete characterisation of uncertainty regarding $\boldsymbol{\vartheta}_0$. We can compute any posterior moment $\mathbb{E}[g(\boldsymbol{\vartheta}) \mid D]$, quantile q_τ such that $\Pr(g(\boldsymbol{\vartheta}) \leq q_\tau \mid D) = \tau$, probability interval $[l_\alpha, u_\alpha]$ such that $\Pr(l_\alpha < g(\boldsymbol{\vartheta}) \leq u_\alpha \mid D) = \alpha$, etc.

Example 5.2.6. Given all available information, ϑ_j is between $l_{0.95}$ and $u_{0.95}$ with 95% probability.

Remark. The interpretation is totally different from CIs.

The distribution $(\boldsymbol{\vartheta} \mid D)$ may be viewed as updating the prior with observed D ; this updating is coherent:

$$f(\boldsymbol{\vartheta} \mid Y_1 = y_1, Y_2 = y_2) = \frac{f(\boldsymbol{\vartheta}) f(y_1, y_2 \mid \boldsymbol{\vartheta})}{f(y_1, y_2)} = \frac{f(\boldsymbol{\vartheta} \mid Y_1 = y_1) f(y_2 \mid Y_1 = y_1, \boldsymbol{\vartheta})}{f(y_2 \mid Y_1 = y_1)}.$$

MLE:

1. Econometrician provides information regarding the distribution $(\mathbf{y}, \mathbf{X})$, up to unknown $\boldsymbol{\vartheta}_0$.
2. Econometrician provides information regarding the distribution

$$(\{Y_{n+i}, \mathbf{x}_{n+i}\}_{i=1}^{\infty} \mid \mathbf{y}, \mathbf{X}),$$

up to unknown $\boldsymbol{\vartheta}_0$.

3. Data provides information regarding the unknown $\boldsymbol{\vartheta}_0$.

Posterior:

1. Econometrician provides information regarding the distribution $(\mathbf{y}, \mathbf{X})$, up to unknown $\boldsymbol{\vartheta}_0$.
2. Econometrician provides information regarding the unknown $\boldsymbol{\vartheta}_0$ from prior sources.

3. Data provides information regarding the unknown $\boldsymbol{\vartheta}_0$.

Remark. (i) One should not misunderstand this as "Bayesian is more subjective, while MLE is more objective." Both sides introduce information beyond the data.

(ii) That is, ϑ_0 does not change, the matching with the economic model does not change, and the family does not change. The only thing that changes is the link:

$$\text{distribution} \rightarrow \text{estimator} \rightarrow \text{CI};$$

now it is

$$\text{distribution} \rightarrow \text{prior} + \text{likelihood} \rightarrow \text{posterior}.$$

5.3 Bayesian Normal Linear Regression

Consider the conditionally normal linear regression (LR):

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + U_i, \quad (\mathbf{U} \mid \mathbf{X}) \sim \mathcal{N}(\mathbf{0}, \sigma_u^2 \mathbf{I}_n).$$

Recall that conditional normality implies A1, A4 and A5. Assumptions A2 and A3 are also not necessary in the Bayesian framework.

The Bayesian likelihood is obtained by evaluating the PDF:

$$f(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma_u^2) = (2\pi\sigma_u^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma_u^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right\},$$

at the realisation D .

Assume the prior density is

$$f(\boldsymbol{\beta}, \sigma_u^2) = f(\sigma_u^2) f(\boldsymbol{\beta} \mid \sigma_u^2),$$

where:

$$f(\sigma_u^2) = \frac{(s^2)^\nu}{\Gamma(\nu)} (\sigma_u^2)^{-\nu-1} \exp\left\{-\frac{s^2}{\sigma_u^2}\right\} \quad (\text{Inverse-Gamma}),$$

$$f(\boldsymbol{\beta} \mid \sigma_u^2) = (2\pi\sigma_u^2)^{-\frac{k}{2}} |\mathbf{V}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\sigma_u^2}(\boldsymbol{\beta} - \mathbf{b})' \mathbf{V}^{-1}(\boldsymbol{\beta} - \mathbf{b})\right\} \quad (\text{Normal}).$$

The specification implies the marginal prior distribution for β_j , $j = 1, \dots, k$ is

$$\sqrt{\frac{\nu}{s^2 V_{jj}}} (\beta_j - b_j) \sim t_{2\nu}.$$

$\nu, s^2, \mathbf{b}, \mathbf{V}$ are prior tuning parameters. They are related to prior moments:

$$\mathbb{E}[\sigma_u^2] = \frac{s^2}{\nu - 1}, \quad \text{V}(\sigma_u^2) = \frac{\mathbb{E}[\sigma_u^2]}{(\nu - 2)^2}, \quad \mathbb{E}[\beta_j] = b_j, \quad \text{V}(\beta_j) = \mathbb{E}[\sigma_u^2] V_{jj}.$$

Remark. The truly stronger assumption is not whether one uses inverse-gamma or gamma, but rather this conditioning. The posterior can then be derived directly. This kind of prior family is called conjugate prior family.

The posterior density is:

$$f(\hat{\sigma}_u^2 \mid \mathbf{y}, \mathbf{X}) = \frac{(\bar{s}^2)^{\bar{\nu}}}{\Gamma(\bar{\nu})} (\hat{\sigma}_u^2)^{-\bar{\nu}-1} \exp\left\{-\frac{\bar{s}^2}{\hat{\sigma}_u^2}\right\} \quad (\text{Inverse-Gamma}),$$

$$f(\hat{\boldsymbol{\beta}} \mid \hat{\sigma}_u^2, \mathbf{y}, \mathbf{X}) = (2\pi\hat{\sigma}_u^2)^{-\frac{k}{2}} |\bar{\mathbf{V}}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\hat{\sigma}_u^2} (\hat{\boldsymbol{\beta}} - \bar{\mathbf{b}})' \bar{\mathbf{V}}^{-1} (\hat{\boldsymbol{\beta}} - \bar{\mathbf{b}})\right\} \quad (\text{Normal}),$$

where

$$\bar{\nu} = \nu + \frac{n}{2}, \quad \bar{s}^2 = s^2 + \frac{1}{2}(\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{I}_n + \mathbf{X}\mathbf{V}\mathbf{X}')^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b}),$$

$$\bar{\mathbf{V}} = (\mathbf{V}^{-1} + \mathbf{X}'\mathbf{X})^{-1}, \quad \bar{\mathbf{b}} = \bar{\mathbf{V}}(\mathbf{V}^{-1}\mathbf{b} + \mathbf{X}'\mathbf{y}).$$

Remark. Same form, just update the tuning parameters.

The marginal likelihood is

$$f(\mathbf{y} \mid \mathbf{X}) = (2\pi)^{-\frac{n}{2}} \frac{\Gamma(\bar{\nu}) |\bar{\mathbf{V}}|^{\frac{1}{2}} (\bar{s}^2)^{-\bar{\nu}}}{\Gamma(\nu) |\mathbf{V}|^{\frac{1}{2}} (s^2)^{-\nu}}.$$

5.4 Monte Carlo Experiment

Perform an experiment by generating (pseudo-)random numbers on a computer.

Set “true” data generating process (DGP) to be a conditionally normal LR with

$$k = 3, \quad \boldsymbol{\beta} = (0, -1, 1)', \quad \sigma_u^2 = 0.5.$$

Set the prior tuning parameters:

$$\nu = 5, \quad s^2 = 1, \quad \mathbf{b} = \mathbf{0}, \quad \mathbf{V} = \mathbf{I}_3.$$

Compare the prior, posterior, likelihood and MLE-derived 95% confidence intervals for β_3 from fictitious data sets with

$$n = 5, \quad n = 30, \quad n = 100, \quad n = 1000.$$

Remark. (i) In small samples, the posterior, the likelihood, and the MLE/OLS interval can differ a lot, because both the method and the external information matter.

- (ii) The posterior contains more information than a confidence interval, because it not only tells you which values are reasonable, but also tells you which values inside the interval are more likely.
- (iii) As the information brought by independent samples increases, the posterior moves toward the likelihood, frequentist intervals (OLS, MLE) also shrink, and the differences across methods gradually disappear.

6 Parameter Identification and Instrumental Variable Estimation

6.1 Statistical Identification

Recall:

- A statistical model is a distribution with PDF $f(\mathbf{y}, \mathbf{X})$ such that the data $\mathbf{D}$ is one realisation of $\mathbf{y}$ and $\mathbf{X}$.
- For a parametric family $\mathcal{F}_\vartheta$, the PDF is $f(\mathbf{y}, \mathbf{X}; \vartheta)$, where f is fixed and differences in distribution are attributed entirely to different values of ϑ .

Definition 6.1.1. If two or more PDFs evaluated at $\mathbf{D}$ attain the maximum, the distributions are called observationally equivalent.

Definition 6.1.2. All possible values of ϑ are called a parameter space, denoted Θ ; e.g., $\Theta = \mathbb{R}^k$.

Also recall that we match an “ideal” distribution with PDF $f(\mathbf{y}, \mathbf{X}; \vartheta_0)$ to economic equilibrium models, and

$$\vartheta_0 = \arg \max_{\vartheta} \mathbb{E}[\ln f(\mathbf{y}, \mathbf{X}; \vartheta)].$$

The ideal distribution is observationally equivalent to another distribution if $L(\vartheta_0; \mathbf{D}) = L(\vartheta; \mathbf{D})$ for some $\vartheta \in \Theta$.

Definition 6.1.3 (Identification of a parameter vector). The parameter vector ϑ_0 is identified if for any realisation, there is no other observationally equivalent parameter in Θ . This is equivalent to:

$$f(\mathbf{y}, \mathbf{X}; \vartheta_0) = f(\mathbf{y}, \mathbf{X}; \vartheta) \iff \vartheta_0 = \vartheta.$$

Remark. Lack of identification is not an algorithmic problem; it is an information problem. The issue is the data themselves do not contain enough information to distinguish them.

All estimators (OLS, MLE, etc.) require that ϑ_0 is identified. However, a posterior distribution does not require full identification.

Remark. (i) Without identification, it is difficult to establish the usual asymptotic theory, for example, to justify asymptotic normality or confidence intervals. In other words, an estimate may still be computable, but inference will run into problems.

(ii) Whatever’s in the posterior will just always be replicated from whatever’s in the prior

Example 6.1.4 (OLS identification via a rank condition). Consider

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + U_i, \quad \mathbb{E}[U_i \mid \mathbf{x}_i] = 0, \quad V(U_i \mid \mathbf{X}) = \sigma_u^2.$$

The parameter space is $\mathbb{R}^k \times \mathbb{R}_+$.

Since $\mathbb{E}[U_i \mid \mathbf{x}_i] = 0 \Rightarrow \mathbb{E}[\mathbf{x}_i U_i] = 0$, $\boldsymbol{\beta}$ is identified if

$$\mathbb{E}[\mathbf{x}_i (Y_i - \mathbf{x}_i' \boldsymbol{\beta}_0)] = \mathbb{E}[\mathbf{x}_i (Y_i - \mathbf{x}_i' \boldsymbol{\beta}_1)] \Rightarrow \boldsymbol{\beta}_0 = \boldsymbol{\beta}_1.$$

Rearranging:

$$\mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] (\boldsymbol{\beta}_0 - \boldsymbol{\beta}_1) = \mathbf{0}.$$

So $\beta_0 = \beta_1$ if $\text{rank } \mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] = k$, since then

$$\mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] (\beta_0 - \beta_1) = \mathbf{0} \Rightarrow \beta_0 = \beta_1.$$

Example 6.1.5 (Failure of identification under perfect collinearity). Consider

$$Y_i = \alpha_0 + \alpha_1 X_i + \alpha_2 (0.5 X_i) + U_i, \quad \mathbb{E}[U_i | \mathbf{x}_i] = 0, \quad V(U_i | \mathbf{X}) = \sigma_u^2,$$

with $\mathbf{x}_i = (1, X_i, 0.5 X_i)'$. Then

$$\mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] = \begin{pmatrix} 1 & \mathbb{E}[X_i] & 0.5 \mathbb{E}[X_i] \\ \mathbb{E}[X_i] & \mathbb{E}[X_i^2] & 0.5 \mathbb{E}[X_i^2] \\ 0.5 \mathbb{E}[X_i] & 0.5 \mathbb{E}[X_i^2] & 0.25 \mathbb{E}[X_i^2] \end{pmatrix}.$$

Since the third row is $0.5 \times$ the second row, $\text{rank } \mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] = 2$. Hence $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \alpha_2)'$ is not identified.

However, set

$$\beta_0 = \alpha_0, \quad \beta_1 = \alpha_1 + 0.5 \alpha_2.$$

Then $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ is identified.

In general, if $\boldsymbol{\vartheta}_0$ is not identified, there exists a dimension-reducing transformation of $\boldsymbol{\vartheta}$ that is.

6.2 Structural Identification

Not only must the statistical parameter be identifiable, but one must also ask whether the statistical coefficient can map uniquely to the structural parameter of economic theory.

In the BLP example, we matched a linear regression to the demand equations:

$$\ln(s_i/s_0) = D_{mi}(\{\ln p_i\}, \{\tilde{\mathbf{x}}_i\}, \{\xi_i\}, (\boldsymbol{\phi}', \alpha_m)'), \quad i = 1, \dots, N,$$

by applying a first-order approximation. The regression coefficients $\boldsymbol{\beta}$ were matched exactly to structural parameters $\boldsymbol{\delta}$, interpreted as elasticities. In this case, $\boldsymbol{\beta}$ identified $\Rightarrow \boldsymbol{\delta}$ identified: we can compute $\boldsymbol{\delta}$ precisely from $\boldsymbol{\beta}$.

Remark. It corresponds to the case of exogenous prices and perfect competition. That is, the economic interpretation of the unilateral regression is clean, but the applicable economic context is relatively narrow.

However, in BLP models with monopolistic competition, there is also a set of supply equations:

$$\ln p_i = S_{mi}(\{\ln s_i\}, \{\tilde{\mathbf{x}}_i\}, \{\xi_i\}, \{\omega_i\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)'), \quad i = 1, \dots, N.$$

Remark. The gain from doing so is that your interpretation of elasticity becomes more robust: it is no longer restricted to perfect competition, but is instead placed in a market structure where prices are determined endogenously.

To match a statistical model to BLPs with monopolistic competition, we can apply a first-order approximation of the supply equation and expand the linear regression to a system of linear regressions:

$$Y_{i1} + \beta_{12} Y_{i2} + \mathbf{x}_i' \boldsymbol{\gamma}_1 = U_{i1},$$

$$Y_{i2} + \beta_{21}Y_{i1} + \mathbf{x}'_i\boldsymbol{\gamma}_2 = U_{i2},$$

where $\mathbb{E}[U_{i1} | \mathbf{X}] = \mathbb{E}[U_{i2} | \mathbf{X}] = 0$.

Remark. Once the supply side is introduced, price becomes endogenous. Now price is jointly determined with shares, utility shocks, and marginal costs. Therefore, the residual and price no longer satisfy the standard independence or conditional mean independence assumptions from introductory econometrics. Instead, the residual will contain part of the price determination equation.

The system of linear regressions is called a linear structural equations model (SEM). The term structural comes from the assumption that β_{12} , β_{21} , $\boldsymbol{\gamma}_1$ and $\boldsymbol{\gamma}_2$ can be matched precisely to parameters in the first-order approximation of the BLP equilibrium conditions.

Structural identification: two important questions.

1. Are linear SEM parameters identified from the data?

The real difficulty lies in question 1: this system cannot be estimated directly by OLS.

2. Are BLP parameters identified from linear SEM parameters?

If these SEM coefficients are assumed to map directly to the parameters in the linearized equilibrium, then as long as these SEM coefficients are identified, elasticity is also identified.

The G -equation SEM is written as

$$\mathbf{y}'_i\mathbf{B} + \mathbf{x}'_i\boldsymbol{\Gamma} = \mathbf{u}'_i,$$

where $\mathbf{y}_i, \mathbf{B}, \mathbf{x}_i, \boldsymbol{\Gamma}, \mathbf{u}_i$ have dimensions

$$(G \times 1), (G \times G), (k \times 1), (k \times G), (G \times 1),$$

respectively.

Other notation (same model):

$$\mathbf{B}\mathbf{y}_i + \boldsymbol{\Gamma}\mathbf{x}_i = \mathbf{u}_i, \quad (G \times G)(G \times 1) + (G \times k)(k \times 1) = (G \times 1),$$

and for the g -th equation,

$$\mathbf{y}'_i\boldsymbol{\beta}_g + \mathbf{x}'_i\boldsymbol{\Gamma}_g = U_{ig}, \quad (1 \times G)(G \times 1) + (1 \times k)(k \times 1) = (1 \times 1),$$

where $\boldsymbol{\beta}_g$ and $\boldsymbol{\Gamma}_g$ denote the g -th columns of $\mathbf{B}$ and $\boldsymbol{\Gamma}$.

Set

$$\mathbf{Y}_{(n \times G)} = \begin{pmatrix} \mathbf{y}'_1 \\ \vdots \\ \mathbf{y}'_n \end{pmatrix}, \quad \mathbf{X}_{(n \times k)} = \begin{pmatrix} \mathbf{x}'_1 \\ \vdots \\ \mathbf{x}'_n \end{pmatrix}, \quad \mathbf{U}_{(n \times G)} = \begin{pmatrix} \mathbf{u}'_1 \\ \vdots \\ \mathbf{u}'_n \end{pmatrix}.$$

The entire system may be written $\mathbf{Y}\mathbf{B} + \mathbf{X}\boldsymbol{\Gamma} = \mathbf{U}$ and the statistical model has PDF $f(\mathbf{Y}, \mathbf{X}; \vartheta_0)$.

Assumption 6.2.1. • SEM1: $\mathbf{B}$ is nonsingular and has rank G .

- SEM2: $\text{rank } \mathbb{E}[\mathbf{x}_i\mathbf{x}'_i] = k$.
- SEM3: $\mathbb{E}[\mathbf{u}_i | \mathbf{x}_i] = \mathbf{0}$ and $\mathbb{E}[\mathbf{u}_i\mathbf{u}'_i | \mathbf{X}] = \boldsymbol{\Sigma}$ is a symmetric positive definite matrix.

- SEM4: $\mathbb{E}[\mathbf{u}_i \mathbf{u}_j' \mid \mathbf{X}] = \mathbf{0}$ for all $i \neq j$.

Assumption SEM1 implies we can solve for all endogenous variables in terms of all exogenous variables:

$$\mathbf{y}_i' + \mathbf{x}_i' \Gamma \mathbf{B}^{-1} = \mathbf{u}_i' \mathbf{B}^{-1}.$$

Equivalently,

$$\mathbf{y}_i' = \mathbf{x}_i' \Pi + \mathbf{v}_i', \quad \Pi = -\Gamma \mathbf{B}^{-1}, \quad \mathbf{v}_i' = \mathbf{u}_i' \mathbf{B}^{-1}.$$

This is called the reduced form of the linear SEM. $\mathbb{E}[\mathbf{v}_i \mid \mathbf{x}_i] = \mathbf{0}$ and

$$\mathbb{E}[\mathbf{v}_i \mathbf{v}_i' \mid \mathbf{X}] = (\mathbf{B}^{-1})' \Sigma \mathbf{B}^{-1} \equiv \Omega.$$

Π and Ω are the reduced form parameters.

Under assumptions SEM2–SEM4, the reduced form parameters are identified; e.g.,

$$\Pi = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{y}_i'].$$

The structural form parameters are identified if there is a unique $(\mathbf{B}, \Gamma, \Sigma)$ such that

$$\Pi = -\Gamma \mathbf{B}^{-1} \quad \text{and} \quad \Omega = (\mathbf{B}^{-1})' \Sigma \mathbf{B}^{-1}.$$

Remark. The data can directly provide information about Π and Ω , that is, the reduced-form objects. But if one wants to reverse engineer β , Γ , and Σ , one immediately faces a problem with more unknowns than available information. Therefore, restrictions must be imposed.

Number of parameters must match.

- Reduced form: exactly $Gk + \frac{1}{2}G(G+1)$.
- Structural form: up to $G^2 + Gk + \frac{1}{2}G(G+1)$.

Identification of structural form parameters requires restrictions. Interpretation: insufficient information in the data about all structural form parameters.

We introduce information from an external source about some parameters to allow the data to provide information about other parameters. External source is economic theory: match $\mathbf{B}$ and Σ to structural parameters in economic models to elicit restrictions.

6.3 Introduction to Instrumental Variables

Suppose we are only interested in the structural form parameters of one equation (e.g., demand elasticities):

$$Y_{i1} + \beta_{12} Y_{i2} + \mathbf{z}_{i1}' \boldsymbol{\gamma}_1 = U_{i1},$$

where we match Y_{i1} to $\ln(s_i/s_0)$ and Y_{i2} to $\ln p_i$.

This regression cannot be implemented with OLS if the supply equation is:

$$Y_{i2} + \beta_{21} Y_{i1} + \mathbf{z}_{i2}' \boldsymbol{\gamma}_2 = U_{i2},$$

with $\mathbb{E}[U_{i1} U_{i2} \mid \mathbf{z}_{i1}, \mathbf{z}_{i2}] = 0$ and $\beta_{21} \neq 0$.

The reduced form of the supply equation is

$$Y_{i2} = \mathbf{z}'_i \boldsymbol{\pi}_2 + V_{i2}, \quad V_{i2} = \frac{U_{i2} - \beta_{21} U_{i1}}{1 - \beta_{12} \beta_{21}}.$$

Hence,

$$\mathbb{E}[U_{i1} V_{i2} \mid \mathbf{z}_i] \neq 0 \quad \text{and} \quad \mathbb{E}[U_{i1} Y_{i2} \mid \mathbf{z}_i] \neq 0.$$

Option 1: Estimate the linear SEM in reduced form, impose restrictions on $(\mathbf{B}, \boldsymbol{\Gamma}, \boldsymbol{\Sigma})$ and derive the structural form. This will produce inference on both the demand and supply elasticities.

Option 2: Suppose $\mathbf{z}_i$ is $k_z \times 1$ and set $\mathbf{x}_i = (\mathbf{z}'_{1i}, Y_{i2})'$, which is $k_x \times 1$. Assume $\text{rank } \mathbb{E}[\mathbf{x}_i \mathbf{z}'_i] = k_x$, then use the estimator:

$$\hat{\boldsymbol{\beta}}_{2\text{SLS}} = \left(\mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{X} \right)^{-1} \left(\mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{y} \right).$$

Moreover, $\text{rank } \mathbb{E}[\mathbf{x}_i \mathbf{z}'_i] = k_x$ implies $k_z \geq k_x$; if $k_z = k_x$ then

$$\hat{\boldsymbol{\beta}}_{2\text{SLS}} \equiv \hat{\boldsymbol{\beta}}_{\text{IV}} = (\mathbf{Z}' \mathbf{X})^{-1} \mathbf{Z}' \mathbf{y}.$$

$\hat{\boldsymbol{\beta}}_{2\text{SLS}}$ is called the two-stage least squares estimator and $\hat{\boldsymbol{\beta}}_{\text{IV}}$ is called the instrumental variables estimator.

$\mathbf{z}_i$ is a vector of instrumental variables (IV) for Y_{i2} . It contains all exogenous variables in the linear SEM. Key features:

- $\mathbf{z}_i$ is uncorrelated with U_{i1} [exogenous IVs]
- $\mathbf{z}_i$ is correlated with Y_{i2} (unless $\boldsymbol{\gamma}_2 = \mathbf{0}$). [relevant IVs]

$k_z \geq k_x$ implies $\mathbf{z}_{i2}$ contains at least one exogenous variable that is not in $\mathbf{z}_{1i}$. This corresponds to an exclusion restriction on $\boldsymbol{\Gamma}$ in the linear SEM.

Remark. IV conditions are exactly restrictions on gamma

It is the key restriction that allows for partial identification: β_{21} and $\boldsymbol{\gamma}_1$ are identified, even if β_{21} and $\boldsymbol{\gamma}_2$ might not be.

6.4 The Generalized Method of Moments (GMM)

In many economic models, the equilibrium characterising system of equations

$$F_m(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \{\varepsilon_i\}, \boldsymbol{\theta}) = \mathbf{0},$$

can be uniquely solved for ε_i :

$$\varepsilon_i = h_{mi}(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \boldsymbol{\theta}), \quad i = 1, \dots, n.$$

Then, independence of $\{\varepsilon_i\}$ and $\{\tilde{\mathbf{x}}_i\}$ implies that for $\mathbf{z}_i = \zeta(\tilde{\mathbf{x}}_i)$,

$$\mathbb{E}[h_{mi}(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \boldsymbol{\theta}) \mathbf{z}'_i] = \mathbf{0}, \quad i = 1, \dots, n.$$

Extending to $i = 1, \dots, N$ and applying the WLLN yields:

$$\frac{1}{N} \sum_{i=1}^N h_{mi}(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \boldsymbol{\theta}) \mathbf{z}'_i \xrightarrow{p} \mathbb{E}[h_{mi}(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \boldsymbol{\theta}) \mathbf{z}'_i] = \mathbf{0}.$$

Remark. Here $\boldsymbol{\theta}$ are structural parameters.

This motivates constructing a loss function as follows. Let

$$\mu_{ij}(\boldsymbol{\theta}) = h_{mi}(\{\mathbf{w}_i\}, \{\tilde{\mathbf{x}}_i\}, \boldsymbol{\theta}) z_{ij},$$

and set

$$\mathbf{M}_i(\boldsymbol{\theta}) = (\mu_{i1}(\boldsymbol{\theta})', \dots, \mu_{ik_z}(\boldsymbol{\theta})')'.$$

Note that $\mathbf{M}_i(\boldsymbol{\theta})$ is a $k_\varepsilon k_z \times 1$ vector whose elements depend on $\{\mathbf{w}_i\}$, $\{\tilde{\mathbf{x}}_i\}$ and $\boldsymbol{\theta}$.

Next, set the loss function to be:

$$S_n(\boldsymbol{\theta}) = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{M}_i(\boldsymbol{\theta}) \right)' \mathbf{W}_n \left(\frac{1}{n} \sum_{i=1}^n \mathbf{M}_i(\boldsymbol{\theta}) \right),$$

where $\mathbf{W}_n$ is a full rank, square, symmetric weighting matrix.

In the spirit of M-estimation, the GMM estimator is derived as:

$$\hat{\boldsymbol{\theta}}_{\text{GMM}} = \arg \min_{\boldsymbol{\theta}} S_n(\boldsymbol{\theta}).$$

The properties of $\hat{\boldsymbol{\theta}}_{\text{GMM}}$ are analogous to the M-estimator.

In this formulation, GMM is an approach to estimate the structural parameters $\boldsymbol{\theta}$ directly. This is the most common use of GMM in econometrics; it is equivalent to specifying a distribution with $f(\mathbf{Y}, \mathbf{X}; \vartheta_0)$ where:

- $\vartheta_0 = \boldsymbol{\theta}$;
- a specific economic theory informs certain moments;
- the PDF form f is not fully specified.

Remark. What you are really doing is exchanging robustness to distributional assumptions for dependence on the structural theory of the model; there is no free lunch.

Example 6.4.1. Let

$$h_{mi1}(Y_{i1}, Y_{i2}, \mathbf{z}_{i1}, \beta_{21}, \boldsymbol{\gamma}_1) = Y_{i1} + \beta_{21} Y_{i2} + \mathbf{z}_{i1}' \boldsymbol{\gamma}_1 = Y_{i1} - \mathbf{x}_{i1}' \boldsymbol{\beta},$$

and $\mathbf{z}_i = (\mathbf{z}_{i1}', \mathbf{z}_{i2}')'$ as in the 2SLS/IV example. Then,

$$S_n(\boldsymbol{\theta}) = \left(\frac{1}{n} \sum_{i=1}^n (Y_{i1} - \mathbf{x}_{i1}' \boldsymbol{\beta}) \right)' \mathbf{W}_n \left(\frac{1}{n} \sum_{i=1}^n (Y_{i1} - \mathbf{x}_{i1}' \boldsymbol{\beta}) \right),$$

and

$$\hat{\boldsymbol{\beta}}_{\text{GMM}} = (\mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{X})^{-1} (\mathbf{X}' \mathbf{Z} \mathbf{W}_n \mathbf{Z}' \mathbf{y}).$$

If $\mathbf{W}_n = \sigma^2 (\mathbf{Z}' \mathbf{Z})^{-1}$, then $\hat{\boldsymbol{\beta}}_{\text{GMM}} = \hat{\boldsymbol{\beta}}_{\text{2SLS}}$.

Remark. (i) What we are really doing is taking the residual from the demand equation, treating it as the h function in GMM, and then using the same instruments as in IV to construct the moment conditions. Therefore, the loss function looks exactly like GMM, and the estimate obtained is exactly the same.

- (ii) The other is that, more generally, GMM is more flexible than 2SLS. If one changes the weighting matrix, or uses other instruments, one obtains a slightly different estimator; under suitable conditions, however, they all converge to the same target.

7 Inference from Data I: Hypothesis Testing

7.1 Hypotheses

Hypothesis tests attempt to assess whether there is evidence contrary to a proposed restriction.

Let $\boldsymbol{\psi} = r(\boldsymbol{\vartheta})$ be a $q \times 1$ vector representing some transformation of a $k \times 1$ vector of model parameters $\boldsymbol{\vartheta}$.

Example 7.1.1 (Linear regression examples of $r(\boldsymbol{\vartheta})$). Consider the linear regression model

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + U_i,$$

and set, for example:

- $\boldsymbol{\psi} = r(\beta_1, \beta_2, \beta_3, \sigma_u^2) = \beta_1$;
- $\boldsymbol{\psi} = r(\beta_1, \beta_2, \beta_3, \sigma_u^2) = \beta_2 - \beta_1$;
- $\boldsymbol{\psi} = r(\beta_1, \beta_2, \beta_3, \sigma_u^2) = (\beta_2 - \beta_1, \beta_3 - \beta_2)'$, etc.

Definition 7.1.2. A point hypothesis concerning $\boldsymbol{\psi}$ is a proposed restriction such as $\boldsymbol{\psi} = \boldsymbol{\psi}_0$ where $\boldsymbol{\psi}_0$ is a hypothesized value chosen by the researcher.

- The null hypothesis is $H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0$.
- The alternative hypothesis is $H_1 : \boldsymbol{\psi} \neq \boldsymbol{\psi}_0$.

Definition 7.1.3. The parameter space Θ is the union of parameter values that satisfy the restriction in H_0 and those that do not (equivalently those in H_1):

$$\Theta = \{\boldsymbol{\vartheta} \in \Theta : r(\boldsymbol{\vartheta}) = \boldsymbol{\psi}_0\} \cup \{\boldsymbol{\vartheta} \in \Theta : r(\boldsymbol{\vartheta}) \neq \boldsymbol{\psi}_0\}.$$

In hypothesis testing, every parameter value in Θ either satisfies H_0 or H_1 . Equivalently, H_0 and H_1 split the family of distributions $\mathcal{F}_{\boldsymbol{\vartheta}}$ into distributions satisfying H_0 and distributions satisfying H_1 .

The goal is to assess whether or not the restriction in H_0 is compatible with the observed data:

- Is the realisation $\mathbf{D}$ incompatible with the hypothesis, in the sense that $\mathbf{D}$ would be highly unlikely if the hypothesis were true?
- Equivalently: is $\mathbf{D}$ too extreme to be generated by the distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ that satisfy H_0 ?

7.2 Decision Rule and Test Statistics

A hypothesis test may either Reject H_0 or Fail to reject H_0 . The decision is based on a realisation of random variables, so it is a mapping from the sample space to the decision set.

Definition 7.2.1. It is convenient to express this mapping as a real-valued function of random variables called a test statistic:

$$T(\mathbf{Y}, \mathbf{X}, \psi_0),$$

e.g., t -statistic, F -statistic, etc. Evaluating at the realisation $\mathbf{D}$ yields a realisation of the test statistic

$$\tau = T(\mathbf{D}, \psi_0).$$

Definition 7.2.2 (Decision rule via critical value). The hypothesis test then consists of the decision rule. Given a critical value c :

- Reject H_0 if $\tau > c$.
- Fail to reject H_0 if $\tau \leq c$.

Designing a hypothesis test typically boils down to designing an appropriate test statistic.

If the test statistic has a known distribution, then c is selected such that the distributions in $\mathcal{F}_\theta$ satisfying H_0 yield

$$Pr_0(T(\mathbf{Y}, \mathbf{X}, \psi_0) > c) = \alpha,$$

where α is a pre-determined value called the significance level.

To compute c given α , we must know the CDF of $T(\mathbf{Y}, \mathbf{X}, \psi_0)$:

- This CDF is derived from distributions in $\mathcal{F}_\theta$ satisfying H_0 .
- This CDF cannot depend on any unknown quantities.
- When the CDF is fully known, $T(\cdot)$ is called pivotal.

Example 7.2.3 (LR and OLS). Test statistics in practice are typically simple transformations of estimators.

Recall the OLS estimator $\hat{\beta}$, $\hat{\sigma}_y^2$, and define

$$\hat{t}(\psi_0) = \sqrt{\frac{n-k}{n}} \frac{\hat{\beta}_j - \psi_0}{\widehat{\text{se}}(\hat{\beta}_j)}.$$

Then, the test statistic for $H_0 : \beta_j = \psi_0$ is

$$T(\mathbf{Y}, \mathbf{X}, \psi_0) = |\hat{t}(\psi_0)|.$$

If A6(i) holds, then distributions in $\mathcal{F}_\theta$ satisfying H_0 have PDFs

$$f(\mathbf{y}, \mathbf{X}; \beta_1, \dots, \beta_{j-1}, \psi_0, \beta_{j+1}, \dots, \beta_k, \sigma_y^2) = \prod_{i=1}^n \varphi \left(Y_i - \sum_{l \neq j} \beta_l X_{il} - \psi_0 X_{ij}, \sigma_y^2 \right),$$

and

$$\hat{t}(\psi_0) \sim t_{n-k}.$$

Hence,

$$\Pr_0(T(\mathbf{Y}, \mathbf{X}, \psi_0) > c) = 2(1 - F_t(c; n - k)) = \alpha,$$

where $F_t(\cdot)$ is the CDF of the Student- t distribution, so the critical value is

$$c = F_t^{-1}(1 - \alpha/2).$$

Given a realisation of $\hat{\beta}$ and $\hat{\sigma}_y^2$, we can compute $\tau = T(\mathbf{D}, \psi_0)$ for any desired value of ψ_0 .

Suppose $\psi_0 = 0$.

- If $\tau > c$, then $\mathbf{D}$ is an extreme draw from a distribution with PDF

$$f(\mathbf{y}, \mathbf{X}; \beta_1, \dots, \beta_{j-1}, 0, \beta_{j+1}, \dots, \beta_k, \sigma_y^2),$$

which is not compatible with $\mathbf{D}$. Decision: reject the distributions in $\mathcal{F}_\theta$ satisfying H_0 as unreasonable probability models, thereby reducing the set of candidate models in $\mathcal{F}_\theta$.

- If $\tau \leq c$, then $\mathbf{D}$ is a reasonable draw from such a distribution. Decision: do nothing (fail to reject).

7.3 The Significance Level and Power of the Test

As $\alpha \rightarrow 0$, almost every $\mathbf{D}$ will be a reasonable draw; as $\alpha \rightarrow 1$, almost every $\mathbf{D}$ will be an extreme draw.

- For every reject decision based on some α , there exists another $\tilde{\alpha} < \alpha$ that would yield a fail to reject.
- For every fail to reject decision based on some α , there exists another $\tilde{\alpha} > \alpha$ that would yield a reject.

Definition 7.3.1. The significance level is also known as the probability of type I error or the size of the test.

Correct interpretation: if hypothetical repeated samples were drawn independently from distributions in $\mathcal{F}_\theta$ satisfying H_0 , say $\mathbf{D}^{(1)}, \dots, \mathbf{D}^{(D)}$, and the same test was carried out for each $\mathbf{D}^{(d)}$, then we would obtain R rejections, with

$$\frac{R}{D} \rightarrow \alpha \quad \text{as } D \rightarrow \infty.$$

Each rejection is an error because it would conclude that $\mathbf{D}^{(d)}$ is not compatible with the distribution it was sampled from.

There are two common confusions about α :

- Type I error is often interpreted as: “reject H_0 when it is true.” Confusion: H_0 restricts a family of distributions (i.e., probability models), which is fictitious by nature. Therefore, H_0 cannot be literally “true” or “untrue” without specifying a context.
- Type I error is often written as $\Pr(\text{reject } H_0 \mid H_0)$. Confusion: H_0 involves parameters, which are not random variables, so a probability conditional on H_0 is not defined in standard probability theory. Interpreting this requires a specific context.

Definition 7.3.2. Let $\Pr_1(T(\mathbf{Y}, \mathbf{X}, \psi_0) > c) = \pi(\psi)$ denote the probability obtained from a distribution in $\mathcal{F}_\theta$ satisfying H_1 , i.e., with $r(\theta) = \psi \neq \psi_0$. Then, for all $\psi \in \{\psi = r(\theta), \theta \in \Theta : \psi \neq \psi_0\}$, the function $\pi(\psi)$ is called the power function of the test. Power is also known as the probability of avoiding type II error.

The power function measures the test's ability to distinguish distributions in $\mathcal{F}_\theta$: those satisfying H_0 vs those satisfying H_1 .

A well-designed test statistic exhibits increasing power for ψ “farther away” from ψ_0 ; e.g., larger $\|\psi - \psi_0\|$ yields larger $\pi(\psi)$.

Correct interpretation: if repeated samples were drawn independently from a distribution in $\mathcal{F}_\theta$ satisfying H_1 (with $\psi \neq \psi_0$), say $\mathbf{D}^{(1)}, \dots, \mathbf{D}^{(D)}$, and the same test was carried out for each $\mathbf{D}^{(d)}$, then:

- we would obtain R rejections and $D - R$ failures to reject;
- $R/D \rightarrow \pi(\psi)$ as $D \rightarrow \infty$;
- each failure to reject is an error because it fails to distinguish distributions satisfying H_0 from the one that $\mathbf{D}^{(d)}$ was sampled from.

Useful intuition: the choice of c involves a trade-off between size and power. Given a test statistic, increasing the critical value decreases the size, but also decreases the power.

7.4 Approximate Hypothesis Tests

An exact distribution of an estimator can only be derived in special cases (e.g., conditionally normal LR). The same is true for test statistics: $\Pr_0(T(\mathbf{Y}, \mathbf{X}, \psi_0) > c)$ is rarely known.

Most hypothesis tests rely on the approximation:

$$\lim_{N \rightarrow \infty} \Pr_0(T(\tilde{\mathbf{Y}}, \tilde{\mathbf{X}}, \psi_0) > c) = \alpha.$$

Equivalently,

$$T(\tilde{\mathbf{Y}}, \tilde{\mathbf{X}}, \psi_0) \xrightarrow{d} Z,$$

where the distribution of Z has CDF F_Z , and the critical value is computed from $1 - F_Z(c) = \alpha$.

When a test statistic is designed such that F_Z is known (when $\Pr_0(T(\mathbf{Y}, \mathbf{X}, \psi_0) > c)$ is not), it is called an asymptotic test.

Example 7.4.1. In the LR example, replacing A6(i) with A6(ii) yields:

$$\tilde{t}(\psi_0) = \frac{\tilde{\beta}_j - \psi_0}{\text{se}(\tilde{\beta}_j)}, \quad T(\tilde{\mathbf{Y}}, \tilde{\mathbf{X}}, \psi_0) = |\tilde{t}(\psi_0)|.$$

For distributions in $\mathcal{F}_\theta$ satisfying H_0 ,

$$\tilde{t}(\psi_0) \xrightarrow{d} \mathcal{N}(0, 1), \quad T(\tilde{\mathbf{Y}}, \tilde{\mathbf{X}}, \psi_0) \xrightarrow{d} Z,$$

with

$$F_Z(c) = 1 - 2\Phi(c) \quad \Rightarrow \quad c = \Phi^{-1}(1 - \alpha/2),$$

where Φ is the standard normal CDF.

The power function of this test is, for any $\beta_j \neq \psi_0$,

$$\pi(\beta_j) = \Pr_1 \left(\left| \tilde{t}(\beta_j) + \frac{\beta_j - \psi_0}{\tilde{\text{se}}(\tilde{\beta}_j)} \right| > c \right).$$

For any distribution in $\mathcal{F}_\theta$ with $\beta_j \neq \psi_0$, we have $\tilde{t}(\beta_j) \xrightarrow{d} \mathcal{N}(0, 1)$ and $\tilde{\text{se}}(\tilde{\beta}_j) \xrightarrow{p} 0$, so that

$$\lim_{N \rightarrow \infty} \pi(\beta_j) \rightarrow 1.$$

In any asymptotic test:

- $T(\tilde{\mathbf{Y}}, \tilde{\mathbf{X}}, \psi_0)$ is asymptotically pivotal iff F_Z is fully known;
- c is the asymptotic critical value;
- α is the asymptotic size;
- the test is consistent iff $\lim_{N \rightarrow \infty} \pi(\psi) \rightarrow 1$.

7.5 P-values

The binary outcome (Reject H_0 / Fail to reject H_0) does not fully summarize the evidence in the data. Either outcome ignores the magnitude of the realised test statistic τ , which suggests a degree of evidence against H_0 .

Definition 7.5.1. An alternative to the binary outcome based on a pre-determined α is the p-value:

$$\text{p-value} = \Pr_0(T(\mathbf{Y}, \mathbf{X}, \psi_0) > \tau).$$

- The p-value is a decreasing function of τ .
- The p-value is the smallest significance level at which H_0 can be rejected. Since $\tau > c \Leftrightarrow \text{p-value} < \alpha$, reject H_0 at significance level α iff $\text{p-value} < \alpha$.

The p-value is often misinterpreted, misrepresented and used to make unjustified claims. Common misconceptions:

- The p-value does not indicate the size or importance of an economically meaningful effect. In particular, a smaller p-value does not mean a relationship is “more significant”.
- The p-value cannot be used to rank different models or compare families of distributions.
- The p-value is not the probability that “ H_0 is true” or “ H_1 is false”.

However, the p-value may be viewed as a measure of the strength of information against H_0 . It is good empirical practice to compute and report the p-value, which contains more useful information than the binary decision or appended (*), (**), (***).

7.6 Statistical Significance

Testing requires a pre-determined significance level α . The most common practice is to set $\alpha = 5\%$.

- This is a by-product of traditional tables of critical values and statistical software.
- There is no scientific justification for $\alpha = 5\%$.

When a test rejects H_0 at the 5% significance level it is common to say that the realised test statistic is statistically significant.

Confusion: this is often extended to parameters $\boldsymbol{\vartheta}$ with $H_0 : \vartheta_j = 0$:

- If H_0 is rejected, conclude ϑ_j is statistically significant.
- Interpretation requires specific context (often missing). It is often incorrectly interpreted as implying a “substantial” or “important” economic effect.

When approximating a test with an asymptotic distribution of the test statistic, we aim for a consistent test. But if $\tilde{\boldsymbol{\vartheta}} \xrightarrow{P} \boldsymbol{\vartheta}_0$, then for every fixed α , a consistent test is guaranteed to reject as $N \rightarrow \infty$ any H_0 except $H_0 : \boldsymbol{\psi} = r(\boldsymbol{\vartheta}_0)$.

Consequently, the statement “ ϑ_j is statistically significant” does not distinguish between $\vartheta_{0j} = 10^{-100}$ and $\vartheta_{0j} = 10^{100}$. If ϑ_{0j} is matched to a structural θ that is interpreted as an economic relationship, then $\theta = 10^{-100}$ vs $\theta = 10^{100}$ are economically very different, but equally statistically significant.

7.7 Hypothesis Tests vs Confidence Intervals

Recall that in the LR satisfying A6(i), the test statistic for $H_0 : \beta_j = \psi_0$ is $T(\mathbf{Y}, \mathbf{X}, \psi_0) = |\hat{t}(\psi_0)|$.

Recall also that the $(1 - \alpha) \times 100\%$ confidence interval (CI) is

$$\hat{\beta}_j \pm c \sqrt{\frac{n}{n-k}} \widehat{\text{se}}(\hat{\beta}_j), \quad F_t(c) = 1 - \alpha/2.$$

ψ_0 is below the lower bound of the CI if and only if

$$\sqrt{\frac{n-k}{n}} \frac{\hat{\beta}_j - \psi_0}{\widehat{\text{se}}(\hat{\beta}_j)} > c,$$

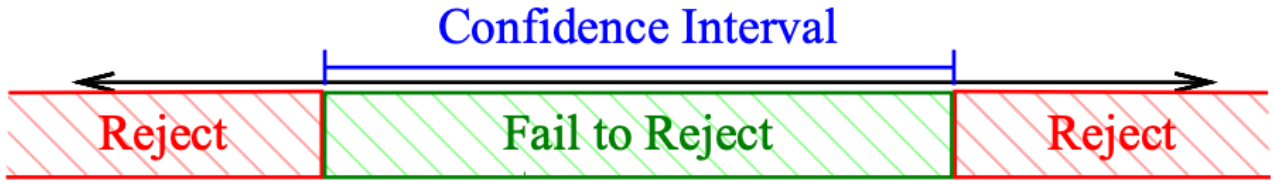
and ψ_0 is above the upper bound of the CI if and only if

$$\sqrt{\frac{n-k}{n}} \frac{\psi_0 - \hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)} > c.$$

Consequently, ψ_0 is outside of the CI if and only if $|\hat{t}(\psi_0)| > c$.

Equivalently, a hypothesis test with $H_0 : \beta_j = \psi_0$ and test statistic $T(\mathbf{Y}, \mathbf{X}, \psi_0)$ will reject H_0 at significance level α for any ψ_0 outside the $(1 - \alpha) \times 100\%$ CI for β_j . This example extends to all hypothesis tests with scalar H_0 (i.e., $q = 1$): the CI for $\boldsymbol{\psi} = r(\boldsymbol{\vartheta})$ contains information on all ψ_0 that would result in rejecting H_0 at significance level α .

The CI supersedes scalar hypothesis tests, including all significance tests.



7.8 Joint Hypothesis Tests

Hypothesis tests only add inferential value when H_0 involves $q > 1$ restrictions.

In principle, we can also construct confidence sets for the $q \times 1$ vector $\psi = r(\vartheta)$. Confidence sets for $q > 1$ adhere to the same principle: any ψ_0 outside the set is rejected as the H_0 , and a H_0 with any ψ_0 inside the set cannot be rejected. Confidence sets for $q > 1$ are extremely rare in practice.

Example 7.8.1. Confidence sets for $q = 2$ are generalised ellipsoids. Confidence sets for $q > 2$ can be derived mathematically, but are difficult to visualise.

Thus, joint hypothesis tests are used to test $H_0 : \psi = \psi_0$ (equivalently, check whether ψ_0 is inside the set).

8 Inference from Data II: Posterior Analysis

8.1 Posterior Inference

Bayesian posterior inference is rooted in simulation. Every draw

$$\vartheta^{o(d)} \sim (\vartheta^o \mid D)$$

yields a possible model

$$f(Y, X; \vartheta_0) \leftrightarrow f(Y, X; \vartheta^{o(d)}).$$

If ϑ_0 is matched to the structural θ of an economic model, then from each $\vartheta^{o(d)}$ we obtain:

- a possible calibration $\theta^{(d)}$ of the economic model;
- possible theory-derived effects;
- possible counterfactuals;
- possible policy interventions;
- possible predictions.

8.2 Probit Example

Recall from Tutorial 4 the analysis of private insurance purchase using the Health and Retirement Study (HRS) data.

- Observe insurance purchase as a binary outcome along with individual characteristics.

- Use Probit regression to obtain inference on average marginal effects and predicted choice probabilities.

We matched the Probit regression to the demand for private insurance utility model:

$$Y_i^* = \mu_u + \tilde{\mathbf{x}}'_{o,i} \boldsymbol{\theta}_o - \sigma_u U_i,$$

such that

$$\Pr(Y_i = 1 \mid \mathbf{x}_i) = \Pr(Y_i^* \geq 0 \mid \mathbf{x}_i) = \Phi(\mathbf{x}_i' \boldsymbol{\beta}).$$

A Bayesian probit regression may be constructed as:

$$\Pr(Y_i = 1 \mid \mathbf{x}_i, \boldsymbol{\beta}^\circ) = \Phi(\mathbf{x}_i' \boldsymbol{\beta}^\circ), \quad \boldsymbol{\beta}^\circ \sim \mathcal{N}(\mathbf{b}, \mathbf{V}).$$

For every posterior draw $\boldsymbol{\beta}^{(d)}$, we obtain a calibrated utility model:

$$\frac{Y_i^*}{\sigma_u} = \mathbf{x}_i' \boldsymbol{\beta}^{(d)} - U_i.$$

We can use posterior draws to simulate:

- (normalised) utilities Y_i^*/σ_u for observed individuals, $i = 1, \dots, n$;
- predicted (normalised) utilities for unobserved individuals, $i > n$;
- predicted choice probabilities $\Pr(Y_i^* \geq 0 \mid \mathbf{x}_i)$ for unobserved individuals, $i > n$.

Set $\mathbf{b} = \mathbf{0}$, $\mathbf{V} = 10\mathbf{I}_k$ and use Gibbs sampling to simulate.

	Mean	Median	Std. Dev.	2.5%	97.5%
retire	0.0419	0.0421	0.0182	0.0058	0.0775
age	-0.0032	-0.0032	0.0024	-0.0081	0.0015
hstatusg	0.0704	0.0703	0.0196	0.0316	0.1083
hhincome	0.0004	0.0004	0.0001	0.0002	0.0007
educyear	0.0250	0.0250	0.0030	0.0193	0.0311
married	0.1294	0.1297	0.0198	0.0905	0.1684
hisp	-0.1675	-0.1668	0.0392	-0.2451	-0.0933

Table 2: Posterior Inference for AMEs

	Estimate	Std. Error	2.5%	97.5%
retire	0.0419	0.0181	0.0064	0.0775
age	-0.0031	0.0024	-0.0079	0.0016
hstatusg	0.0699	0.0194	0.0318	0.1080
hhincome	0.0004	0.0002	0.0001	0.0007
educyear	0.0251	0.0029	0.0194	0.0308
married	0.1267	0.0189	0.0896	0.1638
hisp	-0.1564	0.0328	-0.2207	-0.0922

Table 3: MLE Inference for AMEs

i	retire	age	hstatusg	pincome	educyear	married	hisp
149	0	67	0	0.047	3	0	1
292	0	65	0	0.091	12	0	0
259	1	65	0	0.081	16	0	0

Table 4: Personal Characteristics

i	Mean	Median	Std. Dev.	2.5%	97.5%
149	-1.9788	-1.9423	0.9385	-3.8921	-0.2875
292	-1.1642	-1.0678	0.7554	-2.8621	-0.0697
259	-0.9589	-0.8512	0.6764	-2.5195	-0.0444

Table 5: Posterior: Utilities $\frac{Y_i^*}{\sigma_u}$ of Selected Individuals

8.3 Bayesian Approach to Specification Uncertainty

Recall that a Bayesian model is fully characterised by the joint PDF:

$$f(\boldsymbol{\vartheta}^\circ, Y, X).$$

All specification uncertainty amounts to different specifications of $f(\boldsymbol{\vartheta}^\circ, Y, X)$:

- **Prior uncertainty:** alternative specifications of $f(\boldsymbol{\vartheta}^\circ)$.
- **Likelihood uncertainty:** alternative specifications of $f(Y, X \mid \boldsymbol{\vartheta}^\circ)$.
- **Joint uncertainty:** uncertainty in $f(\boldsymbol{\vartheta}^\circ, Y, X)$ due to prior, likelihood, or both.

In the Bayesian framework, specification uncertainty is treated with random variables.

Consider a set of models $f_m(\boldsymbol{\vartheta}^\circ, Y, X)$, for $m = 1, \dots, M$. Define the discrete RV m° , with support in $\{1, \dots, M\}$.

- $f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ = m) = f_m(\boldsymbol{\vartheta}^\circ, Y, X)$;
- $f(\boldsymbol{\vartheta}^\circ, Y, X) = \sum_{m=1}^M \Pr(m^\circ = m) f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ = m)$.

$\Pr(m^\circ = 1), \dots, \Pr(m^\circ = M)$ are **prior relative model weights**.

- $\Pr(m^\circ = m) \geq 0$ and $\sum_{m=1}^M \Pr(m^\circ = m) = 1$.
- **Intuition:** a prior distribution over candidate models, summarising information about models $m = 1, \dots, M$ available from sources outside the data.

The RV m° is used to implement **Bayesian Model Averaging**.

8.4 Bayesian Model Averaging

The Bayesian model can be redefined so that m° is also a parameter:

$$f(\boldsymbol{\vartheta}^\circ, m^\circ, Y, X) = f(m^\circ) f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ) \equiv \Pr(m^\circ = m) f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ = m).$$

- The **full posterior** is

$$f(\boldsymbol{\vartheta}^\circ, m^\circ \mid D) = \frac{f(\boldsymbol{\vartheta}^\circ, m^\circ, D)}{\int \sum_{m=1}^M \Pr(m^\circ = m) f(\boldsymbol{\vartheta}^\circ, D \mid m^\circ = m) d\boldsymbol{\vartheta}^\circ}.$$

- The **model- m posterior** is

$$f(\boldsymbol{\vartheta}^\circ \mid m^\circ = m, D) = \frac{f_m(\boldsymbol{\vartheta}^\circ, D)}{\int f_m(\boldsymbol{\vartheta}^\circ, D) d\boldsymbol{\vartheta}^\circ}.$$

- The **model-averaged posterior** is

$$f(\boldsymbol{\vartheta}^\circ \mid D) = \sum_{m=1}^M \Pr(m^\circ = m \mid D) f(\boldsymbol{\vartheta}^\circ \mid m^\circ = m, D).$$

$\Pr(m^\circ = 1 \mid D), \dots, \Pr(m^\circ = M \mid D)$ are **posterior model weights**:

$$\Pr(m^\circ = m \mid D) = \frac{\Pr(m^\circ = m) \int f_m(\boldsymbol{\vartheta}^\circ, D) d\boldsymbol{\vartheta}^\circ}{\sum_{m=1}^M \Pr(m^\circ = m) \int f_m(\boldsymbol{\vartheta}^\circ, D) d\boldsymbol{\vartheta}^\circ} = \frac{\Pr(m^\circ = m) f_m(D)}{f(D)}.$$

Here $f_m(D)$ is the model- m marginal likelihood and $f(D)$ is the model-averaged marginal likelihood.

- **Intuition:** a posterior distribution over candidate models, summarising information about $m = 1, \dots, M$ from all sources, including the data.
- Posterior weights summarise all specification uncertainty.

If $\Pr(m^\circ = m) = 1/M$, then

$$\Pr(m^\circ = m \mid D) = \frac{f_m(D)}{\sum_{m=1}^M f_m(D)}.$$

The **Bayes factor** is

$$BF_{l,m} = \frac{f_l(D)}{f_m(D)}.$$

- The evidence in the data in favour of model l relative to m .
- If $\Pr(m^\circ = m) = 1/M$, then $BF_{l,m} = \Pr(m^\circ = l \mid D) / \Pr(m^\circ = m \mid D)$.

8.5 Bayesian Hypothesis Testing

Competing hypotheses can always be formulated in terms of model comparison. For the private insurance example:

- H_0 : (all else equal) education reduces utility from private health insurance;
- H_1 : (all else equal) education increases utility from private health insurance.

In terms of the economic model:

$$H_0 : \theta_{o,5} \leq 0 \quad \text{vs.} \quad H_1 : \theta_{o,5} > 0.$$

In terms of the Probit regression, this translates to:

$$H_0 : \beta_6 \leq 0 \quad \text{vs.} \quad H_1 : \beta_6 > 0.$$

Recall that $H_0 : \beta_6 \leq 0$ vs. $H_1 : \beta_6 > 0$ cannot be compared directly using an MLE-based approach. Instead, we re-formulate:

1. Assume $\beta_6 \geq 0$ (i.e. restrict the parameter space Θ).
2. Test $H_0 : \beta_6 = 0$ vs. $H_1 : \beta_6 > 0$.

The test rejects H_0 if it is sufficiently incompatible with D ; otherwise we fail to reject.

In a Bayesian Probit regression, we formulate H_0 and H_1 as two models which differ in terms of priors.

Let $1_S(x)$ be an indicator function: yields 1 if $x \in S$, 0 otherwise.

- **Model 1:** $f_1(\beta_6^\circ) = 2\phi(\beta_6^\circ; 0, 10) 1_{(-\infty, 0]}(\beta_6^\circ)$.
- **Model 2:** $f_2(\beta_6^\circ) = 2\phi(\beta_6^\circ; 0, 10) 1_{(0, \infty)}(\beta_6^\circ)$.

(Here $\phi(\cdot; 0, 10)$ denotes the $\mathcal{N}(0, 10)$ density.)

The relative evidence in favour of H_0 is $\Pr(m^\circ = 1 \mid D)$; in favour of H_1 is $\Pr(m^\circ = 2 \mid D)$.

Because the half-normal prior densities $f_1(\beta_6^\circ)$ and $f_2(\beta_6^\circ)$ are nested within the normal prior density, we do not need to do anything special to compare the two hypotheses. In particular, if $\Pr(m^\circ = 1) = \Pr(m^\circ = 2) = 0.5$, then

$$\Pr(m^\circ = 1 \mid D) = \Pr(\beta_6^\circ \leq 0 \mid D), \quad \Pr(m^\circ = 2 \mid D) = \Pr(\beta_6^\circ > 0 \mid D).$$

In the HRS example, we estimate $\Pr(\beta_6^\circ > 0 \mid D) = 1$ and conclude evidence overwhelmingly supports H_1 . In fact, we estimate $\Pr(\beta_6^\circ > 0.039 \mid D) = 1$.

8.6 Bayesian Tests of Restrictions

The same approach applies to testing exact restrictions such as $H_0 : \beta_6 = 0$ vs. $H_1 : \beta_6 \neq 0$.

- **Model 1:** prior CDF $F_1(\beta_6^\circ) = 1_{[0, \infty)}(\beta_6^\circ) \Rightarrow \Pr(\beta_6^\circ = 0) = 1$.
- **Model 2:** prior CDF $F_2(\beta_6^\circ) = \Phi(\beta_6^\circ / \sqrt{10}) \Rightarrow \beta_6^\circ \sim \mathcal{N}(0, 10)$.

The model posterior $\Pr(m^\circ = m \mid D)$, $m = 1, 2$, again summarises all evidence regarding H_0 and H_1 .

In this case, the normal prior does not nest Model 1:

$$\beta_6^\circ \sim \mathcal{N}(0, 10) \Rightarrow \Pr(\beta_6^\circ = 0) = 0,$$

which contradicts $F_1(\beta_6^\circ)$.

A prior that does nest Model 1 and Model 2 has CDF:

$$F(\beta_6^\circ) = \Pr(m^\circ = 1) 1_{[0, \infty)}(\beta_6^\circ) + \Pr(m^\circ = 2) \Phi(\beta_6^\circ / \sqrt{10}).$$

In practice, three approaches to testing exact restrictions are:

1. Use the “mixture prior” to derive a nesting posterior.
2. Derive model-specific posteriors individually and compute marginal likelihoods or Bayes factors.

Computing Bayes factors is simplest in Bayesian regressions with natural conjugate priors (Lecture 5 / Tutorial 5). Consider $H_0 : \beta_j = 0$, $H_1 : \beta_j \neq 0$:

- Model 1: regression with X_{ij} excluded;
- Model 2: unrestricted.

In both cases, $f_m(\mathbf{y} \mid \mathbf{X})$ has a simple formula; hence

$$BF_{1,2} = \frac{f_1(\mathbf{y} \mid \mathbf{X})}{f_2(\mathbf{y} \mid \mathbf{X})}$$

is easy to compute without simulation.

3. Derive the Model 2 posterior and compute Bayes factors.

In models with nonconjugate priors, such as the Probit regression, $BF_{1,2}$ can still be computed from the posterior of Model 2 using the Savage–Dickey density ratio (SSDR):

$$BF_{1,2} = \frac{f_{\beta_6^\circ | \mathbf{y}, \mathbf{X}}(0 \mid D)}{f_{\beta_6^\circ}(0)},$$

where $f_{\beta_6^\circ | \mathbf{y}, \mathbf{X}}$ is the marginal posterior and $f_{\beta_6^\circ}$ the prior in Model 2.

Not surprisingly, we find $BF_{1,2} \approx 0 \Rightarrow \Pr(m^\circ = 1 \mid D) \approx 0$; however, restrictions close to the posterior mode are strongly supported.

Restriction	0	0.04	0.06	0.07	0.08	0.10
$BF_{1,2}$	0.00	0.51	169.70	380.89	204.99	0.50
$\Pr(m^\circ = 1 \mid D)$	0.000	0.337	0.994	0.997	0.995	0.334

It is easy to tell which restrictions will be supported—and to what extent—from the simple histogram of posterior draws (e.g. of β_6°): restrictions near the bulk of the histogram (near the posterior mode) tend to be strongly supported; restrictions far in the tails tend to be weakly supported.

8.7 Types of Testable Hypotheses

The Bayesian approach using model posterior distributions is very flexible.

Hypotheses involving parameters, which lead to models with alternative priors, may be:

- **multiple:** e.g. $H_0 : \beta_6 = 0$, $H_1 : \beta_6 = 0.04$, $H_2 : \beta_6 = 0.06$, $H_3 : \beta_6 = 0.07$, $H_4 : \beta_6 = 0.08$, $H_5 : \beta_6 = 0.10$, $H_6 : \beta_6 \in \mathbb{R}$;
- **overlapping:** e.g. $H_0 : \beta_6 \leq 0$, $H_1 : \beta_6 > 0$, $H_2 \in \mathbb{R}$.

Hypotheses may involve features of the likelihood: e.g. $H_0 : U_i$ is normal vs. $H_1 : U_i$ is logistic.

Every hypothesis that can be translated to a specification $f_m(\boldsymbol{\vartheta}^\circ, Y, X)$ may be evaluated using $\Pr(m^\circ = m \mid D)$ relative to the other $M - 1$ hypotheses.